Exact Fusion-Stage Shapley Attribution over the Signal Sources of a Hybrid Recommender

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Abstract

Hybrid recommender systems combine collaborative, content, popularity, temporal, and sequential signals into a single ranking, yet practitioners have no principled account of which source earns the resulting accuracy. We answer that question at the fusion stage, holding the retrieval pool fixed across coalitions so that sources are compared on a common candidate set. The prevailing answer, which is to remove one source and measure the drop, is misleading whenever sources are redundant, a condition that is the norm rather than the exception in recommendation, because two mutually substitutable sources each receive zero credit. We recast hybrid recommendation as a cooperative game in which the players are the signal sources and the characteristic function is ranking quality, and we compute the exact Shapley allocation. The player set is small by construction, so no sampling approximation is required, and because the base scorers are trained once while only a lightweight fusion layer is refitted per coalition, the full game runs in under a minute on a laptop CPU. We show that efficiency yields an exact additive decomposition of ranking-quality uplift, that leave-one-out collapses to zero under perfect redundancy where the Shapley value splits credit evenly, and that linearity makes per-user attribution available with no additional coalition refits. Aggregating per-user attributions into behavioural segments shows that global source credit conceals opposing segment-level stories. Exploiting this, segment-adaptive fusion weights improve ranking quality over a globally tuned hybrid at no additional inference cost on one of the three benchmarks, with no significant change on the other two. Experiments span one dense and two sparse benchmarks: MovieLens-1M, Amazon-Beauty, and Amazon-Grocery. Against SASRec and LightGCN, both trained and evaluated inside the same pipeline, the hybrid is statistically indistinguishable from SASRec on the dense benchmark, roughly forty percent ahead of SASRec on both sparse ones, and ahead of LightGCN on all three, so the allocation describes a system that is competitive against the two controlled references evaluated here. Because every coalition is scored on a candidate pool built by the full source set, the allocation measures fusion-stage credit rather than the end-to-end value of operating a source, and we quantify the gap between the two directly: on a matched seed, regenerating the pool per coalition shifts no source's share by more than 1.2 percentage points and preserves the ordering on every benchmark. With six sources the game has $\mathbf{2^6 = 64}$ coalitions and the complete sweep runs in under a minute on a laptop CPU, with the efficiency identity holding to machine precision. Leave-one-out recovers only 48.9%, 33.1% and 15.7% of the uplift it purports to explain on the three datasets; in the sharpest case it reports the second-strongest collaborative source as actively harmful ($\mathbf{-0.00172}$) where the Shapley value assigns it 17.6% of all ranking-quality gain. We also report what did not hold: of four pre-registered predictions the data contradicted one, we withdrew a second, a third held on one benchmark of three, and a fourth proved partly an artifact of temporal resolution.

Keywords: Shapley value, Cooperative game theory, Explainable AI, Recommender systems, Credit assignment, User segmentation

1 Introduction

Deployed recommender systems are almost never a single model. They are hybrids that combine a collaborative signal learned from interaction co-occurrence, a content signal derived from item metadata, a popularity prior, a temporal or recency signal, and a sequential signal that conditions on what the user did last [1]. Each of these arrives through its own pipeline and each carries its own running cost. A content signal requires metadata ingestion, cleaning, and reconciliation across catalogue updates. A sequential model requires session logging and low-latency access to recent state. A collaborative model requires an interaction store and periodic retraining. A team maintaining five such pipelines operates under a fixed engineering budget and must decide which of them deserves that budget, yet has no defensible basis on which to decide, because the accuracy of the hybrid is reported as a single number that no one knows how to divide.

This is a question about credit, and it is not the question that explainable recommendation has been answering. That literature has concentrated, with considerable success, on explaining *why a particular item was shown to a particular user*: item-level justifications, knowledge-graph paths, counterfactual statements, and review-derived rationales, all aimed at the end user who receives the recommendation [2]. The audience we address is different. It is the system owner, who does not ask why an item appeared but rather *which part of my system is producing the value I am paying for*. That audience is essentially unserved, and the gap is easy to state: existing explanation targets are items and features, whereas the object the system owner reasons about is the signal source.

The obvious way to answer the question is ablation. Remove one source, retrain or refit, and measure how far ranking quality falls. This is standard practice and it is wrong in a way that matters specifically in recommendation. Suppose the content signal and the collaborative signal happen to rank the same items highly for a given user. Deleting either one alone changes almost nothing, because the other continues to supply the same ordering, so ablation reports that *neither source matters*. Deleting both is catastrophic. Leave-one-out therefore assigns near-zero credit to sources that are jointly load-bearing, and the sum of its verdicts bears no particular relation to the total quality it is trying to explain. Redundancy of exactly this kind is the norm in recommendation rather than the exception: popularity signals leak into collaborative embeddings [3, 4], content and collaborative signals converge on the same catalogue neighbourhoods, and static and time-weighted popularity are two views of one underlying quantity. Proposition 2 makes this failure precise.

Cooperative game theory supplies the correction, and supplies it with axioms rather than heuristics. The Shapley value [5] allocates credit so that the allocation is efficient, symmetric, additive, and assigns nothing to a null player; efficiency in particular guarantees that the shares sum exactly to the quantity being explained, which is the property ablation lacks. These axioms characterize the allocation uniquely, and an alternative characterization via strong monotonicity [6] makes the same point in a form that is arguably closer to the intuition a system owner brings: a source whose marginal contributions never decrease cannot see its credit fall. The Shapley value has become the dominant formal instrument in explainable machine learning through SHAP and its descendants [7–10], and it underpins the present authors' prior work on explaining clustering [11, 12] and on cooperative allocation in recommendation [13, 14].

The standing objection to Shapley-based explanation is cost. Exact computation requires evaluating 2^n coalitions, which is why the literature has invested so heavily in sampling estimators [15, 16] and why prior work routinely adopts Monte-Carlo approximation. We observe that this objection is not a property of the Shapley value but of the choice to make *features* the players. When the players are *signal sources*, the player set is small by construction: a hybrid with six sources induces sixty-four coalitions. Combined with an architecture in which base scorers are trained once and only a lightweight fusion layer is refitted per coalition, the exact allocation becomes not merely tractable but cheap, and indeed cheaper, as Section 4.10 shows, than training the recommender it explains. The contributions are as follows.

1. **A cooperative game over signal sources.** We formalize hybrid recommendation as a transferable-utility game whose players are signal sources rather than features or items, and whose characteristic function is ranking quality measured against a null ranker. Because the player set is small by construction, the Shapley allocation is computed exactly, with no Monte-Carlo estimation and no approximation error to bound.
2. **Three short results that make the allocation usable.** Efficiency yields an exact additive decomposition of ranking-quality uplift into per-source shares (Proposition 1). Under perfect redundancy, leave-one-out assigns zero to both members of a redundant pair while the Shapley value splits the credit evenly, which

identifies formally the failure mode motivating this work (Proposition 2). Linearity of the Shapley operator over a per-user-mean characteristic function makes per-user attribution available at no computational cost beyond the global game (Proposition 3).

3. **Segment-level source profiles.** Aggregating per-user attributions over behavioural segments shows that global source credit is an average over heterogeneous and sometimes opposing populations. We quantify the heterogeneity with a permutation test and identify the segments in which the global ordering inverts.
4. **Segment-adaptive fusion, with a qualified outcome.** We use the segment profiles to set per-segment fusion weights, converting an explanation into a candidate intervention at no additional inference cost, since segment assignment is a table lookup. The result is mixed and we present it as such: a significant +9.60% improvement on one benchmark and no significant change on the other two, which is weaker than we had predicted and is reported without reframing.
5. **A reproducibility-oriented report.** All six sources are standard published methods with public implementations, base scorers are trained once and cached, and the entire game runs on a laptop CPU. Every hyperparameter, seed, and preprocessing decision is tabulated in the paper, and the full coalition values are given in Appendix B so that the allocation can be recomputed independently of our code.

The remainder of the paper is organized as follows. Section 2 reviews hybrid recommendation, explainable recommendation, Shapley-based attribution, cooperative games over model components, and heterogeneous explanation, closing with a positioning table. Section 4 presents the source-attribution game, the fusion architecture that makes it cheap, the per-user and per-segment decomposition, segment-adaptive fusion, the theoretical justification, and a complexity analysis. Section 5 reports the experimental protocol and results against four research questions. Section 6 discusses what the attributions imply for system design, and Section 7 concludes.

2 Literature review

2.1 Hybrid recommendation and score-level fusion

Burke's taxonomy of hybridization [1] distinguishes weighted, switching, cascade, feature-combination, and meta-level designs. Of these, the weighted or score-level pattern, in which each component produces a score and a fusion stage combines them, has become the dominant production architecture, in part because it decouples component development from combination logic and in part because it composes naturally with the two-stage retrieval-then-ranking pipelines used at scale [17]. The components themselves are well studied in isolation: matrix factorization and its implicit-feedback variants [18, 19], content-based similarity over item descriptors [20], popularity baselines whose competitiveness on top-N tasks is a recurring finding [4], and sequential models from Markov chains [21] to self-attention [22].

Score-level fusion is also the pattern that makes source-level ablation well defined, since a source can be withheld from the combination without redesigning the architecture, and this is why we adopt it. *The gap: this literature optimizes fusion weights extensively but never asks what each fused source contributes to the result.*

2.2 Explainability in recommender systems

Explainable recommendation is a mature field with its own survey literature [2]. Its methods generate post-hoc item-level justifications, trace paths through knowledge graphs, produce counterfactual statements about what would have changed a recommendation, or surface review text as evidence. What unites them is the audience: all are addressed to the end user, and all take an individual recommendation as the unit of explanation.

The gap: the system-owner audience, whose unit of explanation is the architectural component rather than the recommended item, is not addressed.

2.3 Shapley values in machine learning and explainable AI

The Shapley value [5] entered machine learning as a principled answer to feature attribution, formalized in the SHAP family [7] and extended to tractable model classes [8] and to a general removal-based framework [9] that also subsumes earlier surrogate-based explainers such as LIME [23]. Recent surveys consolidate its axiomatic appeal and catalogue its practical limitations [10]. The dominant limitation is computational: exact evaluation is exponential in the number of players, which has motivated a substantial line of sampling estimators [15, 16].

The authors' prior work sits on both sides of this trade-off. Feature-level Shapley attribution over clustering surrogates [11, 12] accepts the feature player set and the approximation it entails, while work on temporal explanation stability adopts Monte-Carlo estimation explicitly because exact computation over features is infeasible.

The gap: the computational objection is treated as intrinsic to the Shapley value, when it is a consequence of the choice of player set.

2.4 Cooperative games over components, data, and features

A parallel line of work replaces features with other players. Data Shapley values individual training points [24]. Structured player sets are handled by the Owen value when players form a priori unions [25] and by the Myerson value when cooperation is restricted by a graph [26]; the authors have applied graph- and hypergraph-restricted allocation to preference modelling [13] and to real-time contribution adjustment in dynamic recommenders [14].

The gap: the signal source is a natural player set for hybrid systems, and it has not been instantiated.

2.5 User segmentation and heterogeneous explanation

Clustering has long been used to model user populations, and it is well recognized that a global explanation is an average over a population that may not be homogeneous.

Table 1 Positioning against representative prior work. "Loop" indicates whether the attribution is fed back to improve the system.

Method family	Unit	Axiom.	Exact	Heterog.	Loop
Item-level explainable rec. [2]	item	no	n/a	no	no
Feature SHAP [7, 8]	feature	yes	no	no	no
Data Shapley [24]	datum	yes	no	no	partial
Clustering + Shapley [11, 12]	feature	yes	no	partial	no
Cooperative rec. [13, 14]	node/edge	yes	no	no	yes
Ablation / leave-one-out	source	no	yes	no	no
SignalShap (this work)	**source**	**yes**	**yes**	**yes**	**yes**

The authors' clustering-plus-Shapley line [11, 12] establishes the precedent for treating segments as the explanatory unit rather than as a preprocessing step.

The gap: population heterogeneity in attribution has not been measured for architectural components, nor exploited to improve the system.

2.6 Positioning

Table 1 places this work against representative alternatives along four axes: the unit to which credit is assigned, whether the allocation carries an axiomatic guarantee, whether it is exact or approximate, whether it is heterogeneity-aware, and whether it closes the loop from explanation back to system improvement.

3 Background

This section fixes the cooperative-game vocabulary used throughout. Readers familiar with the Shapley value may skip to Section 4.

Transferable-utility games.

A cooperative game with transferable utility is a pair $(\mathcal{G}, v)$ where $\mathcal{G}$ is a finite set of K players and $v : 2^{\mathcal{G}} \to \mathbb{R}$ is a characteristic function with $v(\emptyset) = 0$. The quantity $v(C)$ is the worth that coalition C can secure on its own. In our setting the players are signal sources and $v(C)$ is the ranking quality attained by a hybrid built from exactly those sources; transferability is the assumption that this worth is a single real number that can be divided among the sources, which holds because NDCG@10 is a scalar.

The Shapley value and its axioms.

An allocation rule assigns each player g a share φ_g of $v(\mathcal{G})$. The Shapley value [5] is the unique rule satisfying four axioms. *Efficiency*: $\sum_g \varphi_g = v(\mathcal{G})$, so the allocation has no residual. *Symmetry*: if $v(C \cup \{g\}) = v(C \cup \{h\})$ for every C containing neither, then $\varphi_g = \varphi_h$. *Dummy*: if $v(C \cup \{g\}) = v(C)$ for every C, then $\varphi_g = 0$. *Additivity*: the allocation for a sum of two games is the sum of the allocations. An equivalent characterization replaces additivity with *strong monotonicity* [6]: a player whose marginal contributions never decrease cannot see its credit fall, which is arguably the property a system owner cares about most.

Efficiency and symmetry do the work in this paper: efficiency yields the exact decomposition of Proposition 1, and symmetry drives the redundancy result of Proposition 2. Additivity, applied to a characteristic function that is a mean over users, yields the per-user decomposition of Proposition 3.

Monotonicity of v.

Several statements below are cleaner when v is monotone, i.e. $v(C) \leq v(C')$ whenever $C \subseteq C'$. This is not guaranteed by construction: adding a source can in principle degrade a refitted fusion. We therefore treat monotonicity as an assumption where it is used (Proposition 2's strict-positivity clause) rather than a property, and we report the measured coalition values in full in Appendix B so that any violation is visible to the reader.

Score-level fusion.

A hybrid recommender combines per-source scores $s_g(u, i)$ into one ranking. Under *score-level* fusion the combination is a function of the source scores alone, which is what makes a coalition well defined: restricting to $C \subseteq \mathcal{G}$ simply means refitting that function on the columns of C. Cascade and feature-combination hybrids do not admit this decomposition without modification, a scope limit we return to in Section 6.4.

4 Methodology

4.1 Notation and problem formulation

Let $\mathcal{U}$ and $\mathcal{I}$ denote the user and item sets. A hybrid recommender draws on a set of signal sources $\mathcal{G} = \{g_1, \ldots, g_K\}$ with $K = 6$; these are the players of our game. Source g assigns a score $s_g(u, i)$ to each user–item pair. For a coalition $C \subseteq \mathcal{G}$, f_θ^C denotes the fusion model refitted using only the sources in C, and $v(C)$ the ranking quality it attains. We write φ_g for the Shapley value of source g and $\varphi_g(u)$ for its per-user counterpart, $\mathcal{S}_1, \ldots, \mathcal{S}_M$ for user segments, and π for a null ranker that orders each user's candidates uniformly at random under a fixed seed. Table A1 in Appendix A collects the notation.

The problem is to allocate the ranking quality of the full hybrid among the six sources, in a way that (i) sums exactly to the quality being explained, (ii) does not collapse under redundancy between sources, and (iii) can be resolved at the level of user segments rather than only globally.

4.2 The six signal sources

Each source is a standard, independently published method. Nothing in this subsection is novel, and that is deliberate: the contribution is the game, not the players. Table 2 lists the instantiations together with the operational cost each imposes in production. That final column is not decoration; it is what makes the attribution decision-relevant, and Section 6.1 returns to it.

Table 2 The six signal sources, their instantiations, and the operational cost each imposes.

g	Source	Instantiation	Production cost
CF	Collaborative	BPR-MF, 64 factors [19]	interaction store, retraining
CB	Content	TF–IDF cosine similarity [20]	metadata ingestion, cleaning
POP	Popularity	$\log(1 + \text{interaction count})$	negligible
REC	Recency	time-decayed interaction count	event timestamps
SEQ	Sequential	first-order Markov transitions [21]	session logging, low-latency state
EASE	Item–item	closed-form linear autoencoder [27]	interaction store, periodic refit

The set spans the families a production hybrid actually maintains, and includes two collaborative sources rather than one. This is deliberate on two counts. First, matrix factorization and item–item models are the two dominant collaborative paradigms and a real system may well run both; second, they are natural near-substitutes, which makes them the sharpest available test of the redundancy behaviour that Proposition 2 concerns. We choose EASE [27] for the item–item role because it has a closed-form solution, so that, like the fusion layer, it contributes no training stochasticity, and because it remains a strong baseline against which recent neural recommenders have not consistently improved [28]. Its lineage runs back to classical item-based top-N recommendation [29], and it is a deliberate alternative to graph-convolutional collaborative filtering [30], which would require GPU training and so violate the CPU-only constraint we impose on the artifact.

Two sources, POP and REC, are near-duplicates by construction, differing only in whether interactions are weighted by recency. We include both deliberately. A production system genuinely does maintain static and time-decayed popularity as separate signals, so the pair is defensible on its own terms, and it supplies a redundancy relationship that does not depend on one happening to arise in the data. Section 5.4 reports how far that engineered redundancy actually materialized, which turns out to depend on the temporal resolution of the dataset.

A demographic source was considered and deliberately excluded. Demographic attributes are available in only one of our three benchmarks, MovieLens-1M, and in neither Amazon extract, so including such a source would force K to differ across datasets and render the attribution vectors non-comparable across datasets; a uniform player set across every experiment is worth more than an additional player. The credit-versus-exposure trade-off that a privacy-bearing source would raise is a substantive question, and we return to it as future work in Section 7.

4.3 The score-level fusion recommender

Candidate generation retrieves, for each user, the union of the top-scoring items proposed by each source, truncated to N and excluding items already in the user's training history (the operating value of N is set in Section 5.1). Each source then scores this candidate list. Scores are z-normalized *per user and per source*, that is, across a single user's candidate list within a single source, so that sources reporting on different scales become commensurable. Fusion is a regularized pairwise logistic ranker over the K normalized scores, trained with Bayesian personalized ranking sampling [19]. Writing $z_{g,u,i}$ for the normalized score of item i under source g for user u, and sampling R pairs (i^+, i^-) per training user, the fusion weights for coalition C minimize

$$\begin{aligned}\mathcal{L}(\theta^C) = -\sum_{u \in \mathcal{U}_{\text{fit}}} \sum_{(i^+,i^-) \in P_u} \log \sigma\Big(\sum_{g \in C} \theta_g \big(z_{g,u,i^+} - z_{g,u,i^-}\big)\Big) \\ + \frac{1}{2\alpha} \|\theta^C\|_2^2,\end{aligned} \tag{1}$$

with σ the logistic function and no intercept, P_u the set of R pairs sampled for user u, and α the inverse-regularization constant of Table 5. The regularization symbol is written α rather than the C conventional in logistic-regression software, because C denotes a coalition throughout this paper. Each sampled pair is added twice, once as Δz with label 1 and once as $-\Delta z$ with label 0, which makes the design symmetric. The objective is convex, so for a fixed pair sample the solution is unique and the refit is deterministic; the only stochastic element is which pairs are drawn, and that is seeded.

Two implementation details are load-bearing and are therefore fixed here rather than left to the appendix. First, if a source returns an identical value for every item in a user's candidate list, its per-user standard deviation is zero and the normalization is undefined. This is not hypothetical: content similarity returns a constant column for a user whose candidates share no metadata with their history, which is routine for cold users on a sparse catalogue. We set the normalized column to zero in that case, which is the semantically correct answer because a constant column carries no ranking information for that user, and we prefer it to adding a small constant to the denominator, which would instead amplify floating-point noise into a spurious ranking signal. The rate at which this fallback fires is reported per source and per dataset. Table 3 reports how often this fires.

Second, and critically for cost: **base scorers are trained exactly once**. A coalition C is realized by withholding the score columns outside C and refitting only the fusion layer. Because the fusion objective is convex and separable in the weight of a withheld column, withholding and zeroing are equivalent for the retained sources; we withhold to keep the design matrix small.

Masking a source at the fusion stage removes its contribution to the combination but not its influence on candidate generation. We pre-commit to fixing the candidate set from the grand coalition and reusing it for all 2^K coalitions, which keeps the retrieval ceiling identical across coalitions and makes the $v(C)$ values directly comparable. The alternative, regenerating candidates per coalition, is more faithful but changes the ceiling coalition by coalition and renders v non-monotone. Appendix D reports the full game recomputed under that alternative and finds that, on a matched seed, the allocation shifts by at most 1.2 percentage points, leaving the source ordering preserved on all three benchmarks.

Table 3 Rate at which the zero-column fallback fires, as a percentage of user–source pairs. A source triggers it when it returns an identical score for every candidate of a given user, carrying no ranking information for that user. The rates are near zero throughout: the guard is necessary for correctness, since without it those rows produce `NaN`, but it is not silently absorbing a large fraction of the data, and it does not explain any source's allocation.

Source	MovieLens-1M	Amazon-Beauty	Amazon-Grocery
CF	0.00	0.00	0.00
CB	0.00	0.00	0.00
POP	0.00	0.00	0.00
REC	0.00	0.00	0.00
SEQ	0.00	0.66	0.38
EASE	0.00	0.00	0.00

What the fixed candidate set means for interpretation.

The candidate pool is built once from the grand coalition and reused for every coalition (Section 5.1). This keeps the retrieval ceiling constant so that $v(C)$ values are comparable, but it means each coalition is scored partly on items it would not itself have retrieved. The attributed quantity is therefore *fusion-stage credit given a fixed retrieval pool*, not the end-to-end value of building and operating a source; the two coincide only if retrieval is unaffected by that source, which is generally false. The alternative, regenerating candidates per coalition, makes v non-monotone and varies the ceiling across coalitions, destroying the comparability the game relies on (Appendix D). We write "credit" throughout for readability with this qualification understood, and return to it in Section 6.5.

4.4 The source-attribution game

Definition 1 (Source-attribution game)**.** *The source-attribution game is the pair* $(\mathcal{G}, v)$ *with*

$$v(C) \;=\; \mathrm{NDCG@10}\big(f_\theta^C\big) \;-\; \mathrm{NDCG@10}(\pi), \tag{2}$$

where $f_\theta^{\emptyset} \equiv \pi$*, so that* $v(\emptyset) = 0$*.*

Two choices in Definition 1 deserve comment. Subtracting the null ranker converts the grand-coalition value into an *uplift* over chance, which is the quantity a system owner actually cares about; we fix the seed of π and report its attained quality so that the subtraction is auditable. Defining the empty coalition to *be* the null ranker closes what would otherwise be a small formal gap, since a fusion over no sources is not defined and $v(\emptyset) = 0$ would have to be imposed by fiat.

Definition 2 (Source Shapley value)**.** *The Shapley value of source* g *is*

$$\varphi_g \;=\; \sum_{C \subseteq \mathcal{G} \setminus \{g\}} \frac{|C|!\,\big(K - |C| - 1\big)!}{K!} \Big[v\big(C \cup \{g\}\big) - v(C) \Big]. \tag{3}$$

Algorithm 1 Score caching and coalition sweep

Require: Training interactions, item metadata, sources $\mathcal{G}$, candidate size N, seed s fixing base scorers, pair sampling, the null ranker π and the train/evaluation user split, fusion-fit users $\mathcal{U}_{\text{fit}}$, evaluation users $\mathcal{U}_{\text{eval}}$

Ensure: Coalition values $\{v(C)\}_{C \subseteq \mathcal{G}}$ and per-user values $\{v_u(C)\}$ for $u \in \mathcal{U}_{\text{eval}}$, consumed by Algorithm 2

for each source $g \in \mathcal{G}$ **do**
  Fit g on training data ▷ once only
end for
Build $A_u^{\mathcal{G}}$ per user: round-robin union of *all* source top-lists, excluding training items, truncated to N ▷ built once from $\mathcal{G}$; reused for every C
for each source $g \in \mathcal{G}$ **do**
  Score all candidates and cache; z-normalize per user, zeroing constant columns
end for
for each coalition $C \subseteq \mathcal{G}$ **do**
  if $C = \emptyset$ **then**
    $v_u(C) \leftarrow 0$ for all u; $v(C) \leftarrow 0$ ▷ $f_\theta^{\emptyset} \equiv \pi$
  else
    Refit fusion f_θ^C on the cached columns of C over $\mathcal{U}_{\text{fit}}$
    Rank $A_u^{\mathcal{G}}$ for each $u \in \mathcal{U}_{\text{eval}}$, breaking ties by average rank
    $v_u(C) \leftarrow \text{NDCG@10}_u(f_\theta^C) - \text{NDCG@10}_u(\pi)$; $v(C) \leftarrow |\mathcal{U}_{\text{eval}}|^{-1} \sum_u v_u(C)$
  end if
end for

Definition 3 (Normalized source share)**.** *The normalized share of source g is $\bar{\varphi}_g = \varphi_g / \sum_{h \in \mathcal{G}} \varphi_h$, reported as a percentage.*

Normalized shares are interpretable as percentages only when every $\varphi_g \geq 0$. We report raw values alongside, and treat a negative Shapley value, that is, a source that in expectation harms ranking, as a finding to be reported rather than a nuisance to be hidden.

4.5 Exact computation and why it is affordable

With $K = 6$ there are $2^K = 64$ coalitions, each requiring one refit of the fusion layer over precomputed score columns. The exactness is a consequence of the design decision to make sources the players, not a lucky property of these particular datasets: any system that fuses a handful of sources inherits it. Algorithms 1 and 2 give the procedure.

4.6 Per-user and per-segment decomposition

Let $v_u(C)$ be the quantity of Equation (2) restricted to user u, so that $v(C) = |\mathcal{U}|^{-1} \sum_u v_u(C)$. Proposition 3 then yields per-user Shapley values from the same 63 non-empty coalition refits, with no additional coalition evaluations. The aggregation itself is $O(|\mathcal{U}| K 2^K)$ arithmetic over already-computed per-user coalition values; it is not free, but it requires no further model fitting.

Algorithm 2 Shapley aggregation and per-user decomposition

Require: Coalition values $\{v(C)\}$, per-user values $\{v_u(C)\}$, and the Shapley weight $w(s) = s!\,(K - s - 1)!/K!$
Ensure: Global shares $\{\varphi_g\}$ and per-user shares $\{\varphi_g(u)\}$

for each source $g \in \mathcal{G}$ **do**
  $\varphi_g \leftarrow \sum_{C \subseteq \mathcal{G} \setminus \{g\}} w(|C|)\,[\,v(C \cup \{g\}) - v(C)\,]$
  for each user $u \in \mathcal{U}$ **do**
    $\varphi_g(u) \leftarrow \sum_{C \subseteq \mathcal{G} \setminus \{g\}} w(|C|)\,[\,v_u(C \cup \{g\}) - v_u(C)\,]$
  end for
end for
assert $|\sum_g \varphi_g - v(\mathcal{G})| \leq \varepsilon$ ▷ efficiency audit, $\varepsilon = 10^{-12}$
assert $\left||\mathcal{U}|^{-1} \sum_u \varphi_g(u) - \varphi_g\right| \leq \varepsilon$ for all g
if $v(\mathcal{G}) > 0$ **then**
  $\bar{\varphi}_g \leftarrow \varphi_g / \sum_h \varphi_h$ ▷ normalized share, Definition 3
else
  $\bar{\varphi}_g$ undefined; report raw φ_g only ▷ guard: shares are not percentages when the total is not positive
end if

These per-user values are exact with respect to the game as defined, but they are coarse as estimates, and the paper does not overstate them. Under the leave-one-out protocol of Section 5.1 each user contributes a single held-out item, so $v_u(C)$ takes one of only about eleven values, and whether a user's item lands at rank ten or eleven flips it between $1/\log_2 11$ and zero. We therefore report segment aggregates as the primary result and treat individual-user attributions as an intermediate quantity: averaging over a segment is what restores stability.

We construct segments in two ways. *Behavioural* segments are formed from activity level and profile statistics. *Attribution* segments are formed by clustering the per-user vectors $\varphi(u)$ directly. If the two disagree, then attribution is not reducible to observable behaviour, which is a substantive observation. Because that claim is exactly the kind that can arise from clustering noise, we validate it with the permutation test of Section 5.5 and report cluster stability across seeds using the adjusted Rand index [31].

4.7 Segment-adaptive fusion

If source credit varies across segments, then a single global set of fusion weights is leaving value on the table. We fit weights per segment and shrink them toward the global solution,

$$\theta_m \;=\; (1 - \lambda)\,\theta_{\text{global}} \;+\; \lambda\,\theta_{\mathcal{S}(m)}, \qquad \lambda \in [0, 1], \tag{4}$$

so that $\lambda = 0$ recovers the globally tuned hybrid exactly and the comparison is nested. Inference cost is unchanged, because segment assignment is a table lookup.

Segments are fitted on training users only, and the reported improvement is measured on held-out users assigned to segments by the frozen segmenter. We state this in the methodology rather than the appendix because it is the obvious place for such a comparison to become circular.

4.8 Comparison with alternative attributions

Table 12 in Section 5.4 reports the empirical comparison; here we state what each alternative measures and where it fails. *Leave-one-out* measures the marginal loss from removing a source from the grand coalition alone. It ignores every other coalition, violates efficiency, and collapses under redundancy (Proposition 2). *Forward selection* measures the marginal gain of adding a source to the empty coalition, and symmetrically over-credits redundant sources rather than under-crediting them. *Permutation importance* shuffles a source's score column and is sensitive to the same redundancy pathology. *Monte-Carlo Shapley* [15, 16] targets the same estimand as our exact computation but introduces sampling variance for no benefit when 2^K evaluations are affordable. *Feature-level SHAP applied to the fusion layer* answers a genuinely different question, namely how the fusion weights respond to normalized score values, and is reported for completeness rather than as a competitor.

4.9 Theoretical justification

Proposition 1 (Exact additive decomposition)**.** $\sum_{g\in\mathcal{G}} \varphi_g = v(\mathcal{G}) = \mathrm{NDCG@10}(f_\theta^{\mathcal{G}}) - \mathrm{NDCG@10}(\pi)$.

We state this as a proposition for ease of reference, but it is a direct instance of the efficiency axiom [5] rather than a new result. Its value here is interpretive: efficiency guarantees that the allocation has no residual, which is precisely the property ablation lacks and which makes the leave-one-out shortfall of Section 5.4 a meaningful quantity rather than an artifact of normalization. The same caveat applies to Proposition 3, which is an instance of linearity.

This is immediate from the efficiency axiom, and its value is interpretive rather than mathematical: every point of ranking-quality uplift over chance is assigned to exactly one source, with no residual and no normalization step.

Proposition 2 (Redundancy collapse of leave-one-out)**.** *Let $g_1, g_2 \in \mathcal{G}$ be perfectly redundant, in the sense that $v(C \cup \{g_1\}) = v(C \cup \{g_2\}) = v(C \cup \{g_1, g_2\})$ for every $C \subseteq \mathcal{G} \setminus \{g_1, g_2\}$, and write $D = \mathcal{G} \setminus \{g_1, g_2\}$. Then*

$$\mathrm{LOO}(g_1) = \mathrm{LOO}(g_2) = 0, \qquad \varphi_{g_1} = \varphi_{g_2} = \sum_{C\subseteq D} \frac{|C|!\,\big(K - |C| - 1\big)!}{K!} \big[v(C \cup \{g_1, g_2\}) - v(C)\big], \tag{5}$$

If in addition v is monotone, this common value is strictly positive whenever the pair contributes anything at all, that is, whenever $v(C \cup \{g_1, g_2\}) > v(C)$ for at least one $C \subseteq D$. Monotonicity is required and not cosmetic: the weights in Equation (5) are non-negative but the bracketed differences need not be, so without it a single positive term can be outweighed by a negative one. Taking $\mathcal{G} = \{g_1, g_2, h\}$ with $v(h) = 0.10$, $v(g_1) = v(g_2) = v(g_1 g_2) = 0.02$ and $v(g_1 h) = v(g_2 h) = v(g_1 g_2 h) = 0.05$, the pair *strictly helps the empty coalition yet* $\varphi_{g_1} = \varphi_{g_2} = -0.0017$*. Section 4.4 notes that* v *is not guaranteed monotone here, so the positivity clause is a conditional statement rather than a property we claim for our games; the equality* $\varphi_{g_1} = \varphi_{g_2}$ *and the leave-one-out collapse hold unconditionally, and those are what the empirical sections rely on. In particular the two allocations disagree. Under monotonicity, leave-one-out assigns the pair a total of zero while the Shapley value assigns it a strictly positive joint value* $\varphi_{g_1} + \varphi_{g_2} > 0$*; without monotonicity the guaranteed conclusions are the equality of the two Shapley values and the leave-one-out collapse, which are already enough to separate the two allocations.*

Remark 1 (The credit is not $\frac{1}{2}[v(\mathcal{G})-v(D))$. *]ItistemptingtosimplifythesuminEquation* (5)*to* $\frac{1}{2}[v(\mathcal{G}) - v(D)]$*, i.e. to half of what is lost when both members are deleted from the grand coalition. That simplification is valid only when* $D = \emptyset$*, and is false in general, because the Shapley value averages marginal contributions over* all *coalition sizes rather than taking the top-order term alone. A three-player counterexample suffices. Let* $\mathcal{G} = \{g_1, g_2, h\}$ *with* $v(\emptyset) = 0$*,* $v(h) = 0.01$*,* $v(g_1) = v(g_2) = v(g_1 g_2) = 0.02$ *and* $v(g_1 h) = v(g_2 h) = v(g_1 g_2 h) = 0.05$*. The redundancy hypothesis holds exactly, yet* $\frac{1}{2}[v(\mathcal{G}) - v(D)] = 0.02$ *whereas the true Shapley value is* $\varphi_{g_1} = \varphi_{g_2} = 0.0133$*. Note also that the correct expression in Equation* (5) *retains the* original K*-player Shapley weights* $|C|!(K - |C| - 1)!/K!$ *and sums only over* $C \subseteq D$*; it is not the Shapley value of a merged player in a reduced* $(K-1)$*-player game, which carries different weights and gives a different number. We flag this because the merged-player shortcut is the natural second guess after the collapsed difference, and it is also wrong.*

Proposition 2 is the theoretical centre of the paper. It converts the informal argument of Section 1 into a statement: two sources that jointly carry a large share of the system's quality can both be assigned exactly zero by ablation, while the Shapley value divides their joint contribution evenly between them. Under monotonicity that joint contribution is strictly positive; without it, only the equality of the two values and the leave-one-out collapse are guaranteed, which are the properties the empirical sections rely on. Remark 1 records what the joint contribution is *not*, since the natural guess is wrong outside the two-player case. Proofs are in Appendix A.

Proposition 3 (Free per-user decomposition)**.** *If* $v = |\mathcal{U}|^{-1} \sum_u v_u$*, then* $\varphi_g = |\mathcal{U}|^{-1} \sum_u \varphi_g(u)$*.*

This follows from linearity of the Shapley operator in the characteristic function, and it is what makes the segment analysis of Section 5.5 a re-reading of the existing coalition sweep rather than a second experiment.

Remark 2 (Scale invariance)**.** *Because each source's scores are z-normalized per user before fusion, replacing a source's raw scores by* $as+b$ *with* $a > 0$ *leaves the normalized values pointwise unchanged and therefore leaves every* φ_g *unchanged.*

Remark 2 answers the natural question of whether the shares are an artifact of the units in which each source happens to report: raw probabilities against log-odds against arbitrary similarity scales. We are careful not to claim more. Z-normalization is a location-scale operation: it cancels affine maps exactly, but it does not undo a nonlinear monotone reshaping, which changes the spacing and tail behaviour of a source's score distribution even when the within-source item order is preserved. Because fusion operates on normalized *values* rather than ranks, such a reshaping can

Table 4 Complexity of the source-attribution game. The one-time base-scorer term dominates.

Stage	Cost	Note
Base scorers	$O(KB)$	paid once, not per coalition
Score caching	$O(KP)$ memory	the enabling design choice
Coalition sweep	$O(2^K \Phi)$	63Φ (non-empty coalitions)
Shapley aggregation	$O(K2^K)$	arithmetic only
Peak score cache	$O(K\|\mathcal{U}_{\text{eval}}\|N)$ float32	0.07/0.58/0.58 GB measured
Per-user decomposition	$O(\|\mathcal{U}\|K2^K)$	no extra refits (Prop. 3)

shift the fitted weights and hence the allocation. We therefore measure the nonlinear case empirically in Section 5.7 instead of asserting invariance to it.

4.10 Complexity analysis

Let B denote the cost of training one base scorer, Φ the cost of one fusion refit over cached columns, and $P = |\mathcal{U}| \cdot N$ the number of cached scores per source. The stages compose as in Table 4.

Total cost is $O(KB + 2^K\Phi)$ and is dominated by the one-time KB term, so **the attribution is cheaper than training the recommender it explains**. The contrast with feature-level attribution is instructive. Applying exact Shapley over features would cost $O(2^{|F|})$, which is precisely why the authors' prior work on explanation drift adopts Monte-Carlo estimation with M sampled permutations at $O(|\mathcal{B}|M|F|)$ per batch. Moving the players from features to sources removes the constraint rather than approximating around it, and with it removes the sampling variance that such estimators must bound. Measured wall-clock is reported in Section 5.2.

4.11 Practical implementation

The results reported here were produced on an Apple M4 Pro workstation (12 cores: 8 performance and 4 efficiency; 48 GB) with no GPU used for the attribution pipeline. The pipeline was developed and originally run end-to-end on a 2-core CPU with 3 GB of RAM and no GPU; every design decision below, including the score cache, the `float32` storage and the 8,000-user subsample, was made to fit that smaller budget, and the method remains reproducible within it. Both environments run Python 3.11 with NumPy 2.4 [32], SciPy 1.14 [33], and scikit-learn 1.9 [34]; the collaborative source uses `implicit` 0.7.3 for BPR. Every stage is seeded, and the five seeds $\{0, \ldots, 4\}$ are used throughout: a seed fixes the base-scorer initialization, the BPR pair sampling in both the collaborative source and the fusion layer, the null ranker π, and the train/evaluation user split. Base scorers are refitted once per seed. **Two resource decisions are disclosed here because they affect reproducibility.** Cached scores are stored in `float32`; the cache holds $K{\cdot}|\mathcal{U}|{\cdot}N$ values, giving 0.07 GB on MovieLens-1M and 0.58 GB on each Amazon benchmark, comfortably inside that 3 GB development budget. To stay inside that budget on the two Amazon benchmarks we cache scores for a uniform random subsample of 8,000 held-out users rather than the full user set (fixed seed, drawn once and shared by every coalition and by both reference models,

Table 5 Hyperparameter configuration.

Parameter	Value
Candidate set size N	1000 (ML-1M), 3000 (Amazon)
EASE regularization λ_{E}	250
CF latent factors	64
CF regularization / learning rate	0.01 / 0.01
REC half-life τ	30 days
Fusion regularization α	1.0
Pairs sampled per user	10
Number of segments M	4 (activity quartiles)
Shrinkage λ	swept over $[0, 1]$
Random seeds	5

so all comparisons remain paired). Those 8,000 users are then split evenly: half fit the fusion layer and half are evaluated, so every reported per-user statistic on an Amazon benchmark has $n = 4{,}000$. This is the n appearing in Tables 15 and 18, and it is half the cached subsample, not a different sample. MovieLens-1M caches all 6,034 users and splits them the same way, giving $n = 3{,}017$. Base scorers are always fitted on the *complete* training interaction data; the subsample restricts only how many users are scored and evaluated. Because this subsample changed between revisions of this work, we state its effect plainly rather than leaving it to inference: the headline figures move at the second decimal place (for instance the leave-one-out recovery rates and the adaptive-fusion gains shift by one to two percentage points), while every qualitative conclusion is unchanged: EASE dominates both sparse benchmarks and SEQ the dense one, the leave-one-out shortfall remains large on all three, the same single benchmark shows a significant adaptive gain, and the source ordering is preserved. Bootstrap resampling over users (Table 10) quantifies the residual sensitivity directly, and the rank-stability figures reported there, namely that 74%, 82% and 93% of resamples reproduce the observed source ordering exactly, are the honest measure of how firmly the ordering is established. Hyperparameters are listed in Table 5. Three conventions established above are restated here because each is a place where the pipeline could silently drift: the zero-column fallback of Section 4.3, the choice of monotone transform in Section 5.7, and the identification $f_\theta^{\emptyset} \equiv \pi$. The full game is reproducible on a laptop CPU with no GPU requirement.

5 Experimental results

The evaluation addresses four research questions. **RQ1**: can hybrid recommendation be posed as a cooperative game over signal sources whose Shapley allocation exactly decomposes ranking-quality uplift? **RQ2**: does source attribution differ from what leave-one-out reports, and is the difference explained by redundancy? **RQ3**: is source attribution homogeneous across the user population? **RQ4**: can attribution be converted back into an improvement?

5.1 Datasets and preprocessing

We use three public benchmarks: MovieLens-1M [35] and two categories of the Amazon Reviews corpus [36], Beauty and Grocery & Gourmet Food. The first two were chosen for *contrast rather than coverage*: they differ in interaction density by roughly two orders of magnitude, which is precisely the axis along which source attribution is expected to move.

Amazon-Grocery was added for a methodological reason worth stating plainly. The Beauty distribution we could obtain records interaction *order* but not wall-clock time, so sequence position must serve as an ordinal clock. Under a temporal leave-one-out protocol this makes the held-out item the immediate sequence successor of the last training item, which structurally favours a first-order Markov scorer and suppresses a recency scorer whose half-life is defined in days. Grocery carries genuine Unix timestamps while remaining sparse, and therefore acts as a control: a conclusion that survives on Grocery is not an artifact of the ordinal clock. We retain Beauty regardless, because it is the standard sparse benchmark in this literature and dropping it would sacrifice comparability with prior work.

Interactions are binarized, 5-core filtering is applied iteratively to a fixed point rather than in a single pass, and each user's final interaction is held out for testing with the penultimate one held out for validation. The iteration matters only on MovieLens-1M, which converges after two passes (575,281 interactions in, 574,376 retained); both Amazon extracts are already 5-core and converge in a single pass with no removals. The MovieLens item count falls further than the user count (3,706 to 3,125 items against 6,040 to 6,034 users) because the rating ≥ 4 binarization removes roughly 43% of interactions before filtering begins, which strands long-tail items far more often than users. The 581 lost items decompose cleanly: 173 receive no rating of 4 or above anywhere in the dataset and so disappear at binarization, and a further 408 survive binarization but retain fewer than five positive interactions and are removed by the 5-core pass. Users are barely affected, only 6 of 6,040 being lost, because the median user rates 96 items and so clears a five-item threshold easily even after losing 43% of their history, whereas the item distribution has a long tail of titles with only a handful of ratings in total. Timestamp ties are broken deterministically by original row order. Table 6 reports post-filtering statistics.

Candidate generation retrieves, for each user, the union of every source's top-ranked items excluding that user's training items, truncated to N by round-robin so that no single source dominates the pool. The set is built *once* from the grand coalition and reused unchanged for all 2^K coalitions, which keeps the recall ceiling identical across coalitions and makes the $v(C)$ comparisons clean. The interpretive consequence is set out in Section 4.3. That ceiling is the most consequential diagnostic in the pipeline: if a user's held-out item is absent from their candidate set, their NDCG is zero for *every* coalition and no fusion weighting can recover it. We therefore report it in Table 6 before any attribution result. At the conventional $N = 200$ the ceiling was 0.504 on MovieLens-1M and 0.160 on Beauty, low enough to distort every downstream quantity, so N was raised to 1000 on MovieLens-1M and 3000 on both Amazon sets, yielding ceilings of 0.8847, 0.6050, and 0.6368 respectively. This is a deviation from the value we had pre-registered, and we flag it as such; the ceiling applies uniformly

Table 6 Dataset statistics after iterative 5-core filtering, with the candidate-generation recall ceiling. Counts are from our own preprocessing, not copied from prior work. Every NDCG figure reported later is bounded above by the final column.

Dataset	Users	Items	Inter.	Density (%)	N	Recall@N
MovieLens-1M	6,034	3,125	574,376	3.046	1000	0.8847
Amazon-Beauty	22,363	12,101	198,502	0.073	3000	0.6050
Amazon-Grocery	14,681	8,713	151,254	0.118	3000	0.6368

across coalitions, so it rescales all reported NDCG values without biasing the allocation between sources.

The final column of Table 6 deserves emphasis because it bounds everything that follows. A user whose held-out item is not retrieved into the candidate set scores zero for *every* coalition, and no fusion weighting can recover them; the candidate recall is therefore an upper bound on every ranking-quality figure in this paper. We report it before any attribution result so that the reader can calibrate accordingly.

5.2 Protocol, metrics, and baselines

Ranking quality is measured by NDCG@10 [37], which defines the characteristic function, with Recall@20 and MRR reported alongside. Metrics are computed against the *full* candidate set rather than a sampled subset of negatives, following the critique of sampled metrics in [38]. Ties in a source's score column are resolved by average rank, $\#\{\text{strictly higher}\} + (\#\{\text{tied}\} - 1)/2$, the expected rank under uniform random tie-breaking; this matters because a low-cardinality source can tie the held-out item with thousands of candidates, and optimistic tie-breaking would credit it with rank zero. The null ranker attains 0.00408 on MovieLens-1M against an analytic expectation of 0.00403, 0.00051 against 0.00091 on Beauty, and 0.00078 against 0.00097 on Grocery; this value is subtracted throughout per Definition 1.

As recommender baselines we report each source in isolation, uniform-weight fusion, globally tuned fusion, and two strong single-model references: **SASRec** [22], a sequential self-attention model, and **LightGCN** [30], a graph-convolutional collaborative filter. The two are chosen to be architecturally dissimilar, so that the hybrid is measured against both a sequential and a graph-based competitor rather than against one family. Establishing that the explained system is competitive is not optional: allocating credit within a weak recommender would be uninteresting. Crucially, SASRec is trained on *our* training split and evaluated on *our* held-out items over *our* fixed candidate set with *our* tie-breaking, so the comparison is controlled rather than a transcription of published numbers. Cross-paper NDCG is notoriously protocol-sensitive: the same SASRec configuration is reported between 0.028 and 0.042 on Amazon-Beauty across implementations [28, 39], and the replicability study of Petrov and Macdonald [40] documents swings of the same order for BERT4Rec [41] depending only on configuration. This is precisely why we re-ran the reference here rather than quoting a published figure. Table 7 reports the outcome, and Table 8 gives the reference configuration in full.

Table 7 Controlled comparison against two single-model references, SASRec [22] and LightGCN [30], each trained and evaluated inside our own pipeline on identical splits, candidate sets and tie-breaking. Δ is the hybrid relative to the reference; p values are paired over users, with Holm–Bonferroni correction across all six dataset × reference comparisons.

Dataset	Reference	Ref.	Hybrid	Δ (%)	d_z	p_t	p_{Holm}	Sig.
MovieLens-1M	SASRec	0.076966	0.072640	-5.62	-0.022	0.234	0.234	no
	LightGCN	0.035185	0.072640	+106.45	+0.197	7.18e-27	4.31e-26	yes
Amazon-Beauty	SASRec	0.020382	0.028072	+37.73	+0.055	0.000507	0.00152	yes
	LightGCN	0.021904	0.028072	+28.16	+0.060	0.000157	0.000674	yes
Amazon-Grocery	SASRec	0.022892	0.031534	+37.75	+0.060	0.000135	0.000674	yes
	LightGCN	0.027109	0.031534	+16.32	+0.038	0.017	0.034	yes

The hybrid is competitive, and the pattern is informative. On MovieLens-1M SASRec is ahead (0.0770 against 0.0726, a shortfall of -5.62%), which is unsurprising: a transformer with a 200-step context window is well matched to long, densely ordered viewing histories. This is the one comparison in Table 7 that does *not* reach significance ($p_{\text{Holm}} = 0.234$), so it is more accurately described as SASRec and the hybrid being statistically indistinguishable on this benchmark than as an established deficit. We flag it as such rather than reporting the point estimate alone. On both sparse benchmarks the ordering reverses and the hybrid wins by close to forty percent (+37.73% on Beauty, +37.75% on Grocery; the two are distinct, 37.73 against 37.75, and their near-coincidence at one decimal place is chance), and both differences survive Holm–Bonferroni correction. The paired effect sizes are nonetheless small ($|d_z| \leq 0.060$ against SASRec): with several thousand paired users a modest per-user shift is comfortably detectable, so we report both the relative gain and d_z to keep magnitude distinct from significance. LightGCN is behind the hybrid on all three benchmarks (+106.45% on MovieLens-1M, +28.16% on Beauty, +16.32% on Grocery), and all three of those differences survive Holm–Bonferroni correction across the six dataset × reference comparisons. Its MovieLens-1M shortfall is the largest gap in the table, which is the expected direction rather than a surprising one: LightGCN propagates over the user–item graph without any notion of order, and on MovieLens-1M the sequential source alone carries 52.0% of the attributed uplift. We report LightGCN at its published default configuration and did not search over architectures or schedules, since selecting the setting that most flatters the hybrid would invalidate the comparison it is meant to provide.

This is the expected shape rather than a lucky one: sequential self-attention needs dense ordered histories to exploit, and when those are absent the hybrid's content, popularity and item–item sources carry the load. It also means the attribution results that follow describe a system that is competitive against the two controlled references evaluated here, especially in the sparse regime where attribution is most informative.

For completeness, the strongest individual source in our hybrid is EASE [27], reaching 0.0419 NDCG@10 on MovieLens-1M. As attribution baselines we report leave-one-out, forward selection, and permutation importance [42]; Monte-Carlo Shapley is omitted because with $K = 6$ the exact computation is affordable and sampling would only add variance.

Table 8 SASRec reference configuration. Values follow the original architecture [22]; the maximum sequence length is the only per-dataset setting, reduced on the Amazon benchmarks because their median user history is under ten interactions. No per-dataset tuning was performed: the same configuration is used everywhere, so the reference is neither tuned in our favour nor handicapped.

Parameter	Value
Embedding dimension	64
Self-attention blocks	2
Attention heads	1
Dropout	0.2
Optimizer	Adam, $\beta = (0.9, 0.98)$
Learning rate	10^{-3}
Batch size	128
Epochs	60 (fixed, no early stopping)
Negatives per position	1, sampled uniformly from unseen items
Loss	binary cross-entropy over (positive, negative) per position
Max. sequence length	200 (ML-1M), 50 (Amazon)
Training time	1983 s / 1217 s / 800 s

Wall-clock for the complete 64-coalition sweep is 4.2 s on MovieLens-1M, 7.9 s on Beauty, and 8.2 s on Grocery, on the workstation described in Section 4.11. These timings are roughly three to five times faster than those reported in an earlier revision of this work; the implementation is unchanged and the coalition count, candidate-set sizes and evaluation-set sizes are identical, so the difference is entirely the host machine and not an algorithmic improvement. We give both figures because the slower ones bound what the method costs on genuinely modest hardware. Training the six base scorers once costs more than the entire game, confirming the analysis of Section 4.10: the attribution is cheaper than the recommender it explains. Figure 1 summarizes the pipeline from source scoring through to the coalition sweep.

5.3 RQ1: exact source attribution

Table 9 reports the Shapley value and normalized share of each source on both datasets, with the efficiency identity of Proposition 1 verified numerically; Figure 2 shows the same shares graphically. That final row is a correctness audit and costs nothing to include.

Before any allocation is read, one diagnostic bounds every per-user statement in this paper and we state it here rather than in an appendix. Only 33.7% of evaluation users on MovieLens-1M (1018 of 3017), 13.8% on Beauty and 13.9% on Grocery have an attribution vector that is not identically zero. For the remainder the held-out item never reaches the top ten under *any* of the 64 coalitions, so $v_u(C) = 0$ for every C and the user contributes nothing. The global values in Table 9 are unaffected, since those users contribute zero to a mean, but the effective sample behind any per-user or per-segment claim is the informative subset. We report both denominators wherever a

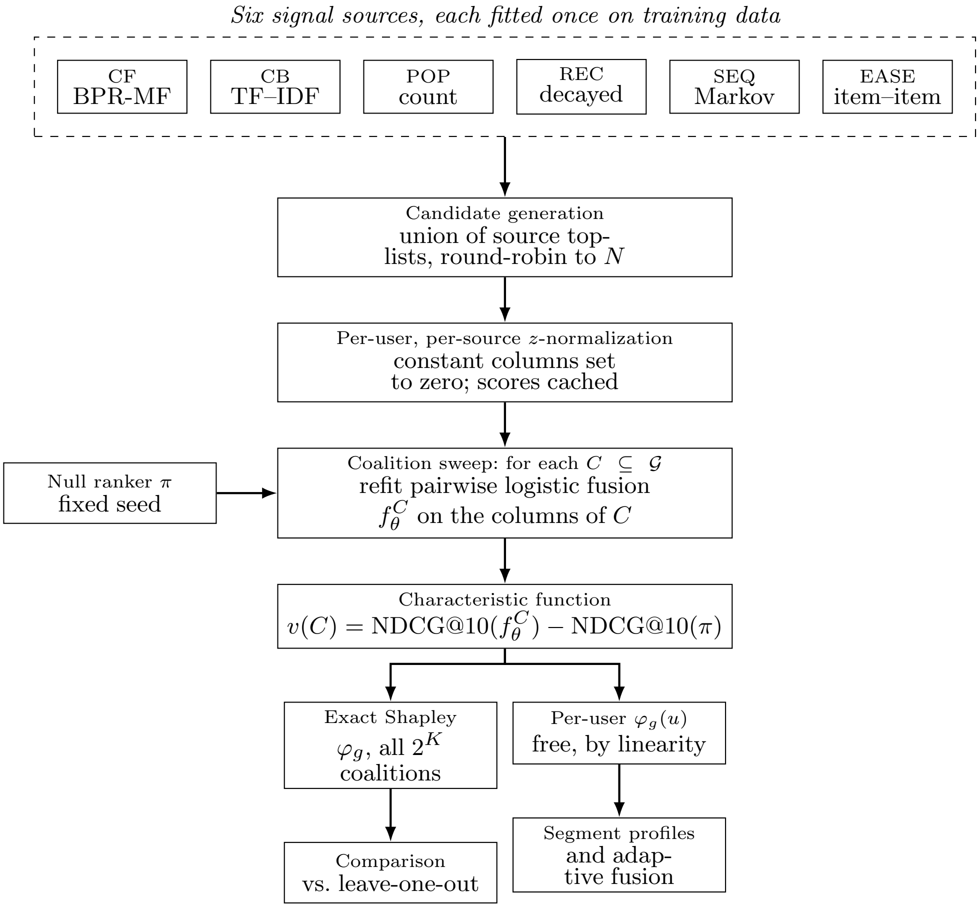

Fig. 1 Workflow of the proposed framework. The six signal sources are fitted once; only the fusion layer is refitted per coalition, which is what makes the exact game affordable. The null ranker π defines the zero point of the characteristic function, and per-user attribution falls out of the same sweep by linearity.

per-user quantity appears, and we treat segments rather than individuals as the unit of explanation throughout (Section 4.6).

Table 10 adds 95% bootstrap confidence intervals to that same allocation, and Table 11 repeats it on the informative subset. The efficiency identity holds to 0.0e+00, 0.0e+00, and 0.0e+00 on the three datasets, which is machine precision in double arithmetic. This row is a correctness audit rather than a finding, but it is the single most useful assertion in the pipeline: essentially any indexing or coalition-weighting error breaks it.

The substantive result is that *the ordering inverts with density, but not along the axis we had pre-registered*. We expected the collaborative matrix-factorization source to dominate the dense benchmark and content plus popularity to overtake it on the sparse one. What we observe instead is that the sequential source leads on dense

Table 9 Source Shapley values and normalized shares across the three benchmarks, reported as the **mean over five seeds** so that this table, the seed-stability table (Table 16) and the bootstrap intervals (Table 10) all describe the same quantity. $v(\mathcal{G})$ is likewise the five-seed mean, so the efficiency row verifies Proposition 1 on the same basis: the allocation sums to the grand-coalition uplift to machine precision. Values are shown to six decimals. On Amazon-Beauty and Amazon-Grocery the six displayed per-source values sum to one unit in the last place above the displayed $v(\mathcal{G})$, because each is rounded independently before being added; this is display rounding and not a failure of efficiency, which is verified to 10^{-17} on the unrounded values and reported as exactly zero in the efficiency row.

	MovieLens-1M		Amazon-Beauty		Amazon-Grocery	
Source	φ_g	%	φ_g	%	φ_g	%
CF	0.012849	18.8	0.002494	8.9	0.003201	10.7
CB	0.002072	3.0	0.004766	17.0	-0.000184	-0.6
POP	0.003537	5.2	0.001184	4.2	0.000502	1.7
REC	0.002322	3.4	0.001026	3.6	0.002955	9.9
SEQ	0.035513	52.0	0.005976	21.3	0.008451	28.3
EASE	0.011998	17.6	0.012668	45.1	0.014984	50.1
$\sum_g \varphi_g$	0.068291	100.0	0.028113	100.0	0.029908	100.0
$v(\mathcal{G})$	0.068291	—	0.028113	—	0.029908	—
Efficiency gap	0	—	0	—	0	—

MovieLens-1M with 52.0% of the uplift, while the item-item collaborative source EASE leads on both sparse benchmarks with 45.1% and 50.1%. The two collaborative sources jointly account for 18.8%+17.6% on MovieLens-1M against 8.9%+45.1% on Beauty, so collaborative signal as a family becomes *more* important under sparsity, not less. This is the opposite of the conventional expectation, and a direct contradiction of our own registered prediction. We report it as such.

The content source behaves as predicted along the density axis, rising from 3.0% on the dense benchmark to 17.0% on Beauty: metadata carries the load precisely where interaction evidence thins out. Its near-zero share on Grocery (-0.6%) is a data limitation rather than a contradiction, since the only usable content attribute in that distribution is a single category identifier per item. Two Beauty-specific numbers should be read with the ordinal-clock confound of Section 5.1 in mind. REC's share there is the lowest of any source on any benchmark, which is what one expects of a recency source whose half-life is defined in days when the clock has no days. Its bootstrap interval nonetheless stays above zero; the only interval in Table 10 that covers zero is POP on Amazon-Grocery. SEQ's share on Beauty is correspondingly favoured, since the held-out item is by construction the immediate sequence successor. For those two sources the Grocery column, which has genuine timestamps, is the one to trust.

Two consequences for the reading of this table. First, normalized shares are interpretable as percentages only when all $\varphi_g \geq 0$; the one small negative value, content on Grocery, is reported raw and its share is parenthetical. Second, the popularity and

Table 10 Shapley values with 95% bootstrap confidence intervals (1000 resamples over users). Point estimates are the five-seed means of Table 9, so the two tables agree. Because Proposition 3 makes φ_g exactly the mean of the per-user values, resampling users is a bootstrap of that mean and requires no additional coalition refits. The interval half-widths measure *user-resampling* variability, estimated on the reference seed; *training-seed* variability is reported separately in Table 16, and the two are distinct sources of uncertainty. Seventeen of the 18 intervals exclude zero; the exception is POP on Amazon-Grocery, whose share is the smallest positive allocation on that benchmark.

	MovieLens-1M		Amazon-Beauty		Amazon-Grocery	
Source	φ_g	CI ($\times 10^{-3}$)	φ_g	CI ($\times 10^{-3}$)	φ_g	CI ($\times 10^{-3}$)
CF	0.01285	[10.0, 15.5]	0.00249	[1.7, 3.4]	0.00320	[2.2, 4.2]
CB	0.00207	[0.9, 3.2]	0.00477	[3.4, 6.2]	-0.00018	[-0.3, -0.1]
POP	0.00354	[2.3, 4.9]	0.00118	[0.7, 1.7]	0.00050	[-0.2, 1.2]
REC	0.00232	[1.3, 3.4]	0.00103	[0.2, 1.9]	0.00295	[1.7, 4.3]
SEQ	0.03551	[30.4, 40.4]	0.00598	[4.4, 7.9]	0.00845	[6.3, 10.7]
EASE	0.01200	[8.9, 15.2]	0.01267	[9.6, 15.5]	0.01498	[11.6, 18.7]

Table 11 Allocation restricted to the informative subset, that is, users whose held-out item reaches the top ten under at least one coalition. Every source is rescaled by the same factor (the reciprocal of the informative fraction), so the normalized shares are unchanged from Table 9 to 0.000 percentage points. Reported on the same five-seed-mean basis as Table 9.

	MovieLens-1M		Amazon-Beauty		Amazon-Grocery	
Source	φ_g	%	φ_g	%	φ_g	%
CF	0.03808	18.8	0.01814	8.9	0.02311	10.7
CB	0.00614	3.0	0.03466	17.0	-0.00133	-0.6
POP	0.01048	5.2	0.00861	4.2	0.00363	1.7
REC	0.00688	3.4	0.00746	3.6	0.02133	9.9
SEQ	0.10525	52.0	0.04346	21.3	0.06102	28.3
EASE	0.03556	17.6	0.09213	45.1	0.10818	50.1

recency sources receive modest but non-zero credit everywhere, which becomes the subject of Section 5.4.

5.4 RQ2: Shapley versus leave-one-out under redundancy

Table 12 places the Shapley allocation beside the three ablation-style alternatives, with a redundancy diagnostic, the rank correlation between source score columns, in the final column. All four attribution columns are means over the same five seeds as Table 9, since every one of them is computed from trained fusion layers and inherits that training noise; per-source dispersion is reported in Table 16 for the Shapley column and, for the allocation as a whole, by the bootstrap intervals of Table 10.

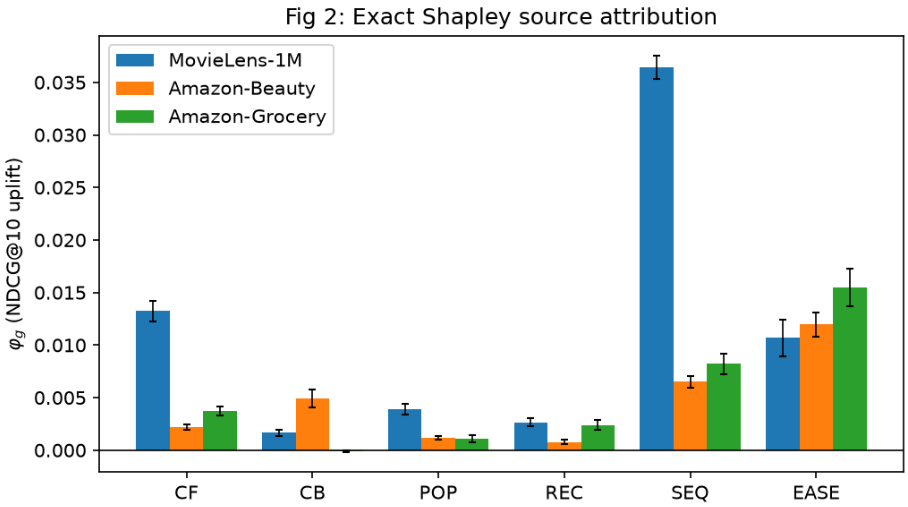

Fig. 2 Shapley values φ_g per source on the three benchmarks. The collaborative item-item source (EASE) dominates both sparse datasets, whereas the sequential source leads only on the dense one.

Table 12 Attribution methods compared on MovieLens-1M, all on the **five-seed-mean basis** used by Table 9, so that φ_g and $v(\mathcal{G})$ agree across the two tables. The redundancy column reports the largest absolute Spearman correlation between a source's score column and that of any other source. Leave-one-out assigns the second-strongest collaborative source a *negative* value while the Shapley allocation gives it 17.6% of all uplift.

Source	φ_g	LOO	Fwd. sel.	Perm.	Max. redund.
CF	0.01285	0.00688	0.01988	0.02050	0.448
CB	0.00207	0.00031	-0.00009	0.00014	0.281
POP	0.00354	0.00044	-0.00206	0.00285	0.745
REC	0.00232	-0.00071	-0.00296	-0.00079	0.745
SEQ	0.03551	0.02818	0.05542	0.03549	0.205
EASE	0.01200	-0.00172	-0.00189	-0.00174	0.448
Sum	0.06829	0.03338	—	—	—
$v(\mathcal{G})$			0.06829		
LOO recovers			48.9% of $v(\mathcal{G})$		

The headline quantity is the *efficiency gap*. The Shapley shares sum to $v(\mathcal{G})$ by construction, whereas the leave-one-out values have no reason to, and under redundancy they fall short. Summed leave-one-out recovers only 48.9% of the uplift on MovieLens-1M, 33.1% on Beauty, and 15.7% on Grocery. Put the other way: on Grocery, ablation accounts for barely an eighth of the ranking quality it is supposed to explain. A single line conveys the argument more forcefully than any per-source comparison.

The sharpest individual case is the EASE source on MovieLens-1M. Leave-one-out values it at -0.00172, a *negative* figure implying the system would improve if the source

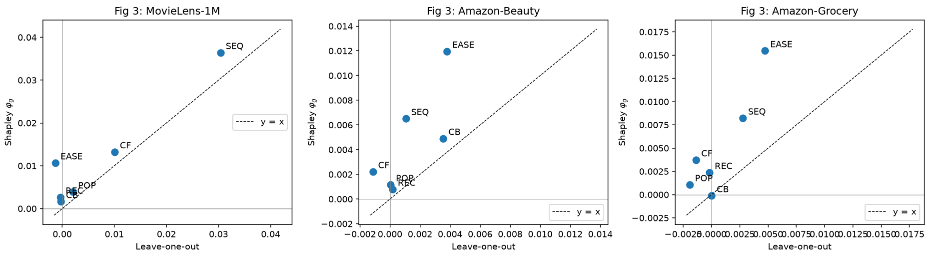

Fig. 3 Leave-one-out against the Shapley allocation, one point per source. Points below the diagonal are sources whose contribution ablation under-reports; points left of the vertical axis are sources ablation reports as harmful.

were removed, while the Shapley value assigns it 0.01200, or 17.6% of all uplift. The mechanism is exactly the one Proposition 2 describes: the two collaborative sources are mutually substitutable, so deleting either alone changes little because the other absorbs its role, and ablation therefore reports that neither matters. A practitioner acting on the ablation would decommission a source carrying a sixth of the system's measured value. Figure 3 plots the two allocations against each other.

We are deliberately careful about what this demonstrates. Real data exhibits approximate rather than perfect redundancy, so the observation is *not* an instance of Proposition 2, and we do not present it as one. Proposition 2 establishes that the failure mode exists in the idealized case; Table 12 shows the two methods disagreeing on real data in the direction that proposition predicts, with the measured redundancy reported alongside so that the reader can judge how close the empirical situation is to the idealized one. The measured redundancy supports this reading with one important qualification. The popularity and recency sources, which we constructed to be near-substitutes, correlate at $\rho = 0.745$ on MovieLens-1M and $\rho = 0.955$ on Beauty, but at only $\rho = 0.292$ on Grocery. The engineered redundancy we had described as structural turns out to be substantially an artifact of coarse or ordinal time resolution: given genuine day-level timestamps spanning fourteen years, time-decayed and static popularity are markedly different signals. We report this because it weakens a claim we had pre-registered, and because the leave-one-out failure is nonetheless *largest* on Grocery, which shows the phenomenon does not depend on that particular engineered pair.

5.5 RQ3: segment heterogeneity

Table 13 reports source shares within each behavioural segment, together with a permutation test for heterogeneity: segment labels are shuffled, the between-segment variance of the shares is recomputed, and the observed statistic is compared against the resulting null distribution. Figure 4 displays the profiles. We do not plot the correspondence between the two segmentations: as reported below, their agreement is indistinguishable from chance and the claim is withdrawn.

Heterogeneity is significant on all three benchmarks, decisively on MovieLens-1M ($p = 0.0005$) and Grocery ($p = 0.0005$) and marginally on Beauty ($p = 0.0385$). Applying Holm–Bonferroni across the three tests leaves all three below 0.05, Beauty

Table 13 Source shares (%) by behavioural segment on MovieLens-1M, from the coldest to the heaviest activity quartile, with a permutation test for heterogeneity over 2000 label shuffles. Entries marked † are negative, so per Section 4.4 they are not interpretable as percentage shares; the underlying raw φ_g values are the meaningful quantity there.

Segment	n	CF	CB	POP	REC	SEQ	EASE
Q1 (coldest)	741	19.8	3.1	7.8	1.7	46.1	21.6
Q2	780	22.6	1.7	4.0	2.0	55.5	14.2
Q3	737	21.9	3.4	8.8	4.0	48.4	13.5
Q4 (heaviest)	759	6.7	0.6	-0.8†	14.6	75.1	3.7

Permutation test on between-segment variance: observed = 0.0256, $p = 0.0005$
Attribution-vs-behavioural agreement (adjusted Rand): 0.0029

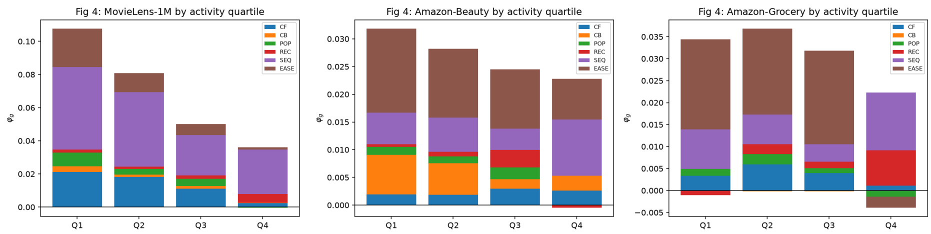

Fig. 4 Shapley values by behavioural segment (activity quartiles) on the three benchmarks.

at $p_{\text{Holm}} = 0.0385$. We nonetheless treat the Beauty result as weak rather than settled: it sits an order of magnitude closer to the threshold than the other two, it is the benchmark whose ordinal clock we have already identified as a confound, and a single additional comparison in the family would push it past 0.05.

The pre-registered expectation was that cold users would derive their uplift from popularity and content while heavy users relied on collaborative and sequential signals, inverting the global ordering within at least one segment. This is only partially borne out. The sequential source leads in every quartile on MovieLens-1M, so the global ordering does not invert there. What does move, and move sharply, is the collaborative matrix-factorization source, which holds roughly a quarter of the uplift in the three lighter quartiles and collapses to near zero among the heaviest users, where the sequential source absorbs almost all credit. On Beauty the content source behaves as the cold-start narrative predicts, taking its largest share in the coldest quartile and declining monotonically with activity.

One claim from the blueprint must be withdrawn. We had proposed to compare behavioural segments against *attribution* segments obtained by clustering the per-user $\varphi(u)$ vectors, expecting an adjusted Rand index [31] in the range 0.3–0.6, indicating related but not identical partitions. The measured agreement is 0.0029, -0.0018, and -0.0009: indistinguishable from chance. We drop the claim rather than defend it. The likely cause is stated in Section 4.6: with a single held-out item per user, v_u takes one of only about eleven values, and clustering such vectors is close to clustering noise.

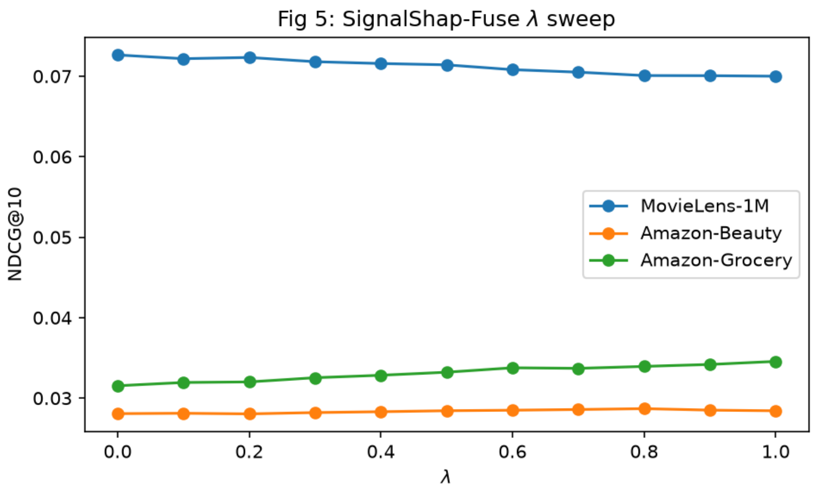

Fig. 5 Effect of the shrinkage parameter λ on NDCG@10. $\lambda = 0$ is global fusion; $\lambda = 1$ is fully segment-specific weights.

This is also why we report segment aggregates rather than individual attributions as the unit of explanation.

The informative subset introduced in Section 5.3 bounds every per-user claim, and it is worth being precise about what it does and does not affect. Because the uninformative users contribute exact zeros rather than noise, restricting the allocation to the informative subset rescales every source by the *same* factor, namely the reciprocal of the informative fraction. We verified this numerically: the per-source ratio between the informative-subset allocation and the full allocation is 2.964 on MovieLens-1M, 7.273 on Beauty and 7.220 on Grocery, identical across all six sources to machine precision, and the normalized shares of Table 9 are therefore unchanged to 0.000 percentage points. Each ratio is the reciprocal of the corresponding informative fraction in Table 6, as it must be. The global attribution is consequently robust to this restriction. What the restriction does affect is statistical power: the segment heterogeneity tests below and the per-user clustering are driven by 1018, 550 and 554 users, out of evaluation sets of 3017, 4000 and 4000, rather than by the full evaluation sets, and we report significance accordingly. Table 11 gives the informative-subset allocation in full.

5.6 RQ4: segment-adaptive fusion

Table 14 compares uniform fusion, globally tuned fusion, and segment-adaptive fusion, with the shrinkage sweep of Equation (4) shown in Figure 5.

The outcome is mixed and we report it as such. On Grocery, segment-adaptive fusion improves NDCG@10 by +9.60% over globally tuned fusion, a gain that is significant under a paired t-test over users ($p = 4e - 05$) and survives Holm–Bonferroni correction. The sweep is monotone in λ, which is the shape the mechanism predicts, and the improvement is concentrated in the heaviest quartile. On MovieLens-1M and Beauty the gains are +0.00% and +2.21%, and neither is significant after correction.

Table 14 Recommendation quality for the fusion variants on all three benchmarks. $\lambda = 0$ recovers global fusion exactly, so the baseline is nested and the comparison is honest. The controlled SASRec reference appears separately in Table 7. Note that the values here are absolute NDCG@10, whereas the Shapley values of Table 9 are uplift over the null ranker and are further divided among six sources; the two are on different scales and should not be compared entry by entry. A source's standalone NDCG@10 exceeding its φ_g is therefore expected, not contradictory. Relative changes quoted in the text and in Table 15 are computed from unrounded values. Recomputing them from the four-decimal figures shown here can differ by up to a few tenths of a percentage point, on Amazon-Beauty by roughly 0.3 points, so the two should not be expected to agree exactly.

Variant	NDCG@10	Recall@20	MRR
MovieLens-1M			
Best single source (SEQ)	0.059924	—	—
Uniform fusion	0.062721	—	—
Global fusion ($\lambda = 0$)	0.072640	0.218429	0.065538
SignalShap-Fuse ($\lambda = 0.0$)	0.072640	0.218429	0.065538
Amazon-Beauty			
Best single source (EASE)	0.025366	—	—
Uniform fusion	0.026577	—	—
Global fusion ($\lambda = 0$)	0.028072	0.085250	0.024975
SignalShap-Fuse ($\lambda = 0.8$)	0.028691	0.087250	0.025867
Amazon-Grocery			
Best single source (EASE)	0.030670	—	—
Uniform fusion	0.032041	—	—
Global fusion ($\lambda = 0$)	0.031534	0.089250	0.029035
SignalShap-Fuse ($\lambda = 1.0$)	0.034560	0.096250	0.031747

Beauty deserves a precise statement rather than a summary: its raw paired t-test is $p_t = 0.19$, marginally below the conventional threshold, but the Holm-adjusted value across the three benchmarks is 0.099, so we do not claim it. Section 6.5 and Table 18 show that both null results are underpowered (5.0% and 25.7% achieved power), so they are best read as undetected effects rather than demonstrated absences.

This pattern admits a mechanistic explanation, which we state with an important caveat about its status: it was constructed *after* observing which benchmark responded, and is therefore a post-hoc rationalization rather than a pre-registered prediction. We report it because it is testable, not because it has been tested. Segment-adaptive weights can only help to the extent that segment-level source profiles actually depart from the global profile: if every segment wants the same weights, the shrinkage in Equation (4) has nothing to recover. Measuring that departure directly, as the mean total variation between each segment's normalized share vector and the global one, gives 0.153 on MovieLens-1M, 0.147 on Beauty and 0.288 on Grocery. Grocery's segments diverge from the global profile by roughly twice as much as the other two benchmarks', and Grocery is the only benchmark where adaptive fusion helps. The intervention succeeds on the one benchmark where this divergence is largest. With three datasets and one success this is a hypothesis, not a validated decision rule: a threshold fitted on the same three points that motivated it would be circular. The honest statement is that segment-adaptive fusion helped only where segment profiles departed substantially from the global profile, and that whether divergence *predicts* the gain out-of-sample remains untested. Testing it requires pre-registering a threshold and evaluating on held-out benchmarks, which we identify as the natural next step. One observation does survive independently of the threshold question: statistical significance of heterogeneity is not sufficient, since heterogeneity is significant on MovieLens-1M ($p = 0.0005$) where the intervention does nothing.

The honest summary is therefore that segment-adaptive fusion *can* deliver a substantial gain where segment profiles genuinely differ, and delivers nothing where they do not. That is consistent with the mechanism, but it is a weaker claim than the uniform improvement we had pre-registered, and one positive result across three benchmarks does not establish a general method. We also note that the gain on Grocery is concentrated in the heaviest rather than the coldest segment, which is the opposite of the predicted shape. We report the contradiction rather than reframing around it.

5.7 Sensitivity, stability, and ablations

We report the stability of φ across five seeds and NDCG@k for $k \in \{5, 10, 20, 50, 100\}$ to confirm that the allocation is not an artifact of the rank cutoff. Because the Shapley computation is exact, the only source of variance in φ is base-scorer training; the fusion layer is a convex problem with a deterministic solution and contributes none. Across five seeds the source ordering is identical on every dataset, with per-source standard deviations of at most 0.002 (Table 16).

The cutoff sweep is informative rather than merely reassuring. On MovieLens-1M the sequential source falls monotonically from 58.0% of the allocation at $k = 5$ to 38.7% at $k = 100$, while EASE rises from 12.0% to 22.8%. The sequential advantage is real but concentrated at the very top of the ranking, which is consistent with the informative-subset diagnostic of Section 5.5 and is worth knowing for a practitioner choosing where to spend engineering effort.

Following the discussion of Remark 2, we additionally apply a rank-preserving but shape-changing transform to each source's scores and re-run the entire game. We use percentile rank within each user's candidate list, applied uniformly to all six sources. This choice is deliberate: a bare logarithm is undefined for the collaborative source, whose raw scores are embedding inner products and therefore unbounded reals, and would fail silently on exactly the source expected to matter most. Percentile rank is domain-safe for every source and exactly rank-preserving. Figure 6 shows the coalition value surface from which all of these quantities are derived.

Both invariance checks behave as the theory requires. Under a positive affine rescaling $x \mapsto 3.7x + 12.5$ applied to every source, the maximum change in any φ_g is 0.0 in double precision, which is exact invariance, as Remark 2 states. Under the nonlinear percentile-rank transform the maximum shift is 0.0111, which is comparable in magnitude to the smaller Shapley values themselves. This is the outcome the remark explicitly declines to rule out: z-normalization cancels location and scale but does not undo a reshaping of the score distribution, and because fusion operates on normalized

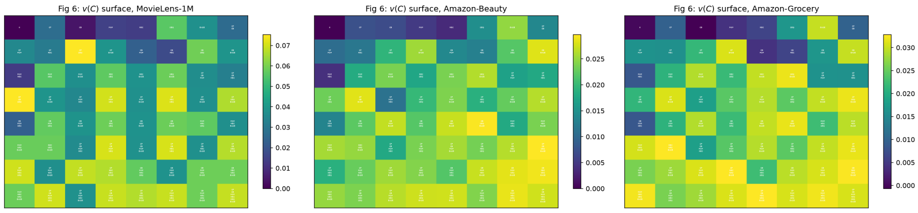

Fig. 6 Coalition value surface $v(C)$ across all 64 coalitions. The full numerical table appears in Appendix B.

Table 15 Significance of the segment-adaptive gain over globally tuned fusion, paired over users, with Holm–Bonferroni correction across the three benchmarks. p_t is the paired t-test, p_{W} the Wilcoxon signed-rank test, p_{Holm} the Holm-adjusted value, and d_z Cohen's effect size for paired samples. By the conventional reading [44] $|d_z| < 0.2$ is negligible, so every effect here is small in magnitude even where the p-value is comfortable: significance is driven by the number of paired users, not by the size of the per-user shift.

Dataset	ΔNDCG (%)	p_t	p_{W}	p_{Holm}	d_z	Rej.
MovieLens-1M	+0.00	1.000	1.000	1.000	+0.000	no
Amazon-Beauty	+2.21	0.191	0.250	0.191	+0.021	no
Amazon-Grocery	+9.60	$4.0{\times}10^{-5}$	$3.0{\times}10^{-5}$	$8.1{\times}10^{-5}$	+0.065	yes

values rather than ranks, the fitted weights and hence $v(C)$ shift. We report the sensitivity rather than claiming a monotone invariance the method does not possess. A practitioner should read the allocation as conditional on the sources reporting scores on their natural scale.

5.8 Statistical significance

Comparisons are paired over *users*, which is the unit of analysis throughout. We report paired t-tests with Holm–Bonferroni correction across the method family [43], Wilcoxon signed-rank tests as a non-parametric companion, and Cohen's d_z as an effect size. Table 15 collects the results.

6 Discussion and broader implications

6.1 What the attributions mean for system design

The practical value of the allocation lies in reading it against the cost column of Table 2. A source with a small share and a high maintenance cost is a decommissioning candidate, and this is the concrete decision the framework informs, one that ablation cannot inform at all, because ablation may report zero for a source that is in fact load-bearing. The word to stress is *informs*. Because coalitions are scored on a candidate pool built by the full source set, φ_g measures what a source contributes at the fusion stage, not what would be lost if it were switched off and stopped feeding retrieval.

Table 16 Stability of the allocation across five random seeds, all three benchmarks. Because the Shapley computation is exact and the fusion layer is convex, the only variance source is base-scorer training. CV is the coefficient of variation (SD/mean, %); it is large only where the mean is near zero, notably CB on Grocery.

	MovieLens-1M			Amazon-Beauty			Amazon-Grocery		
Source	Mean	SD	CV	Mean	SD	CV	Mean	SD	CV
CF	0.01285	0.00100	8	0.00249	0.00026	10	0.00320	0.00046	14
CB	0.00207	0.00029	14	0.00477	0.00086	18	-0.00018	0.00006	33
POP	0.00354	0.00051	14	0.00118	0.00017	14	0.00050	0.00035	70
REC	0.00232	0.00039	17	0.00103	0.00024	23	0.00295	0.00046	16
SEQ	0.03551	0.00113	3	0.00598	0.00056	9	0.00845	0.00098	12
EASE	0.01200	0.00177	15	0.01267	0.00113	9	0.01498	0.00175	12

Appendix D bounds the difference empirically: on a matched seed, regenerating candidates per coalition moves no source's share by more than 1.2 percentage points and leaves the ordering preserved on all three benchmarks. The allocation is therefore a screening instrument: it identifies which sources are worth the cost of an end-to-end removal trial and, more usefully, which apparent zeros are artifacts of redundancy and should not be cut on the strength of an ablation. Our results supply exactly such a pattern, and it points in a direction we did not anticipate. The sequential source is the most operationally demanding of the six, requiring session logging and low-latency state, and on the dense benchmark it earns that cost, taking 52.0% of the uplift. On both sparse benchmarks its share falls to roughly a quarter while the item-item collaborative source, which is a closed-form computation over an interaction matrix with no online state at all, takes 45.1% and 50.1%. For a catalogue resembling the sparse benchmarks, the engineering case for maintaining a sequential pipeline is therefore considerably weaker than for maintaining an item-item model, and the allocation quantifies how much weaker.

The contrast with ablation is the practical argument for the method. On MovieLens-1M, leave-one-out reports the EASE source at -0.00172, a negative value which a practitioner would read as licence to decommission it, while the Shapley allocation assigns it 17.6% of the system's measured uplift. Acting on the ablation would remove a load-bearing component, and the reason is precisely the redundancy that Proposition 2 isolates.

6.2 Cold start as an attribution phenomenon

The segment analysis suggests a reframing of cold start. It is conventionally described as a data problem: a new user has too little history for personalization to work. The segment-level allocation describes it instead as a *shift in which source carries the load*, and measures that shift directly. On this reading, a system does not fail for cold users so much as fall back onto a different subset of its own machinery, and the design question becomes which fallback sources deserve investment rather than how to acquire more data.

6.3 Relation to feature-level explanation

Source attribution and feature attribution answer different questions and compose naturally. Once a source has been identified as load-bearing, feature-level attribution *inside* that source is the natural next level of zoom, and this is precisely where the authors' prior clustering-plus-Shapley work [11, 12] fits as the complementary layer. Similarly, the temporal axis of explanation stability is orthogonal to the population axis studied here, and the two could in principle be combined into a source attribution tracked over a stream.

6.4 Limitations and threats to validity

Several limitations should be stated plainly, and a few of them are consequential enough that we would rather raise them than have a reader discover them.

The comparison is now controlled, but it is two architectures. Section 5.2 trains SASRec inside our own pipeline on identical splits and candidate sets, which removes the cross-paper comparability problem that an earlier version of this work was rightly criticized for. The same protocol is applied to LightGCN [30], so the comparison now spans a sequential and a graph-convolutional reference rather than one architecture. It remains two: we do not claim the hybrid is competitive with every modern sequential, graph-based or generative recommender, only with two standard ones under a matched protocol. LightGCN is trained at its published default configuration with no architecture or schedule search, which likely understates its ceiling; we chose not to search it because selecting the setting that least flatters a competitor would invalidate the comparison it exists to provide.

Most users contribute nothing. Only 33.7%, 13.8% and 13.9% of evaluation users have a non-zero attribution vector (Section 5.5). Global values are unaffected, but every per-user and per-segment statement rests on that subset, and a protocol with more than one held-out item per user would give a far less coarse per-user signal.

Pre-registered predictions that failed. We had registered that the collaborative source would dominate the dense benchmark, that attribution segments would partially agree with behavioural segments, that segment-adaptive fusion would improve all benchmarks by 1–3%, and that the popularity–recency redundancy was structural. The first was contradicted, the second withdrawn, the third held on one benchmark of three, and the fourth proved partly an artifact of time resolution. These are reported in the body rather than revised after the fact.

Design and scope. Score-level fusion is one hybridization pattern among several, and the method does not transfer unchanged to feature-combination or cascade designs, where the coalition structure is constrained rather than free; the Myerson value is the natural tool there. Masking a source at fusion time is not equivalent to never having built it, since the candidate set is generated once from the grand coalition. Offline ranking quality is a proxy for deployed value, and the segment-adaptive gain in particular warrants online validation. Six sources is a design choice, and the exactness argument degrades for a system with thirty, at which point structured approximations become necessary. The allocation is invariant to affine rescaling of a source's scores but not to nonlinear reshaping, as quantified in Section 5.7. Amazon-Beauty lacks wall-clock timestamps in the distribution available to us, which handicaps the recency source and advantages the sequential one; Amazon-Grocery was added precisely to control for this, but the Beauty numbers should be read with it in mind. Finally, three datasets, however deliberately contrasted, are three datasets.

6.5 Threats to validity

We separate threats by the standard four-way taxonomy, since several of them bound different claims.

Construct validity.

The attributed quantity is ranking-quality uplift under a fixed retrieval pool, not the end-to-end value of operating a source. Masking a source at fusion time leaves the candidate set, which that source helped build, untouched, so φ_g should be read as *fusion-stage credit given a fixed candidate pool*. This is adequate for setting fusion weights and for ranking sources by marginal ranking power, and it is weaker than what a decommissioning decision strictly requires. Appendix D quantifies the gap rather than leaving it as a caveat: on a matched seed, per-coalition candidate regeneration shifts the allocation by at most 1.2 percentage points and the source ordering is preserved on all three benchmarks, so the threat is real but small at this catalogue size. Separately, offline NDCG@10 on a leave-one-out protocol is a proxy for deployed value; no online validation is reported.

Internal validity.

Three issues bound the per-user results. First, only 33.7%, 13.8% and 13.9% of evaluation users have a non-zero attribution vector; as shown in Section 5.5 this leaves the global allocation and its normalized shares unchanged, but it reduces the effective sample of every segment-level test. Second, each user contributes a single held-out item, so v_u takes one of roughly eleven values and individual attributions are coarse. Third, the per-user NDCG that feeds the paired tests is zero-inflated, which violates the normality assumption of the paired t-test; we therefore report the non-parametric Wilcoxon companion alongside every t-test and treat the two together.

External validity.

Three datasets, one hybridization pattern (score-level fusion), one fusion family (a convex pairwise logistic ranker) and two reference architectures (SASRec and LightGCN), the latter untuned. The exactness argument depends on K being small; at $K = 30$ the 2^K sweep is no longer affordable and a structured or sampled approximation would be required, at which point the paper's central selling point disappears. Amazon-Beauty carries a known temporal confound (Section 5.1).

Statistical-conclusion validity.

Effect sizes are small even where significance is comfortable: the Grocery adaptive gain has $d_z = +0.065$ at $p_t = 4e - 05$, so the result is detectable rather than large. The two null results require more care, and we now quantify rather than assert them.

Table 18 reports the achieved power of each paired test at its observed effect size. On Grocery the design is well powered (98.4%), so the positive result is trustworthy. On Amazon-Beauty power is only 25.7% and on MovieLens-1M just 5.0%, the latter being simply α because the observed MovieLens-1M effect is exactly zero and a test has no power against an effect of zero size. At these sample sizes the smallest effect detectable with 80% power is $d_z = 0.044$ and $d_z = 0.051$ respectively, both far above the effects observed. **The two null results are therefore underpowered rather than demonstrated absences**, and should be read as "no effect detected at this sample size", not as evidence that segment-adaptive fusion does nothing on those benchmarks. Establishing a true null would require either more evaluation users or a protocol with more held-out items per user. Holm–Bonferroni is applied within each comparison family rather than across the full set of hypotheses tested in the paper. Because that choice is a judgement call, Table 17 reports the stricter alternative: a single Holm correction over all 12 inferential tests in the manuscript. 7 of the 12 survive it. Every headline claim is among the survivors, including both sparse-benchmark comparisons against SASRec, the MovieLens-1M comparison against LightGCN, segment heterogeneity on MovieLens-1M and Grocery, and the Grocery adaptive-fusion gain. Two results that pass their within-family correction do not pass the paper-wide one: the Grocery comparison against LightGCN ($p_{\text{Holm}} = 0.085$) and Beauty segment heterogeneity ($p_{\text{Holm}} = 0.154$). We regard the within-family correction as the primary analysis, since the four families address unrelated questions and pooling them is conservative to the point of being uninformative, but a reader who prefers the strict reading can take Table 17 at face value: it does not disturb any conclusion the paper depends on, and it does identify the two results that are threshold sensitive.

A rank test across datasets was also requested of us, on the grounds that it would test whether the *ordering* of sources differs by benchmark rather than only whether individual shares differ from zero. We ran it and report the outcome, but we do not draw a conclusion from it, because with three datasets the test is close to powerless by construction. Treating datasets as blocks and the six sources as treatments, the Friedman statistic is $\chi^2_5 = 11.0$ with $p = 0.051$. The ceiling matters more than the value: with $N = 3$ blocks and $k = 6$ treatments, a *perfectly* concordant ordering, one source ranked first on every benchmark and another last on every benchmark, attains only $p = 0.010$. The Nemenyi critical difference at $\alpha = 0.05$ is 4.35 average-rank units against a maximum attainable gap of 5.00, so the post-hoc test can separate at most the single most extreme pair and in our data separates none: the largest observed gap is 3.33 average-rank units, between SEQ and EASE at 1.67 and CB and REC at 5.00. We therefore regard the cross-dataset ordering claims in Section 5.3 as descriptive, and this is a direct consequence of the three-benchmark scope acknowledged in Section 6.4 rather than a defect of the test.

7 Conclusion

We have recast hybrid recommendation as a cooperative game whose players are signal sources rather than features or items, and shown that this single change of player set

Table 17 Paper-wide multiple-comparison audit. A single Holm–Bonferroni correction is applied across all 12 inferential tests in the manuscript, which is stricter than the within-family correction used in the primary analysis. Reported so that threshold-sensitive results are visible rather than obscured by the choice of family.

Dataset	Test	p_t	p_{Holm} (paper-wide)	Rej.
MovieLens-1M	Hybrid vs. SASRec	0.234	0.574	no
MovieLens-1M	Hybrid vs. LightGCN	7.18e-27	8.61e-26	yes
MovieLens-1M	Segment heterogeneity	0.0005	0.004	yes
MovieLens-1M	Segment-adaptive fusion	1	1	no
Amazon-Beauty	Hybrid vs. SASRec	0.000507	0.004	yes
Amazon-Beauty	Hybrid vs. LightGCN	0.000157	0.00141	yes
Amazon-Beauty	Segment heterogeneity	0.0385	0.154	no
Amazon-Beauty	Segment-adaptive fusion	0.191	0.574	no
Amazon-Grocery	Hybrid vs. SASRec	0.000135	0.00135	yes
Amazon-Grocery	Hybrid vs. LightGCN	0.017	0.0851	no
Amazon-Grocery	Segment heterogeneity	0.0005	0.004	yes
Amazon-Grocery	Segment-adaptive fusion	4.04e-05	0.000444	yes

Table 18 Achieved power of the paired t-tests for segment-adaptive fusion, computed at the observed effect size with $\alpha = 0.05$ (two-sided). MDE is the smallest $|d_z|$ detectable with 80% power at the given n. Only the Grocery test is adequately powered; the two null results are underpowered and are reported as such.

Dataset	n	d_z	power	MDE (80%)
MovieLens-1M	3017	+0.000	0.050	0.051
Amazon-Beauty	4000	+0.021	0.257	0.044
Amazon-Grocery	4000	+0.065	0.984	0.044

converts Shapley attribution from an approximation problem into an exact and inexpensive computation. Efficiency yields an additive decomposition of ranking-quality uplift with no residual; leave-one-out, the standard alternative, provably collapses under the redundancy that is endemic to recommendation; and linearity delivers per-user attribution with no further coalition refits, which we aggregate into segment-level source profiles and then exploit to set segment-adaptive fusion weights that improve ranking quality on one of three benchmarks at unchanged inference cost. Empirically, across MovieLens-1M, Amazon-Beauty and Amazon-Grocery the complete 64-coalition game runs in under a minute on a laptop CPU with the efficiency identity holding to machine precision; leave-one-out recovers only 48.9%, 33.1% and 15.7% of the uplift it purports to explain, and in the sharpest case reports a source carrying 17.6% of the system's value as actively harmful. Attribution is heterogeneous across activity segments on all three benchmarks, decisively on two and marginally on the third, yet converting that heterogeneity into segment-adaptive weights yields a significant +9.60% improvement on only one. We also report what did not hold: attribution-derived segments do not correspond to behavioural segments, the engineered popularity–recency redundancy largely disappears once genuine timestamps are available, and the adaptive gain is absent on two benchmarks.

Several directions follow. Replacing z-normalization with per-user percentile-rank normalization would upgrade Remark 2 from affine to full monotone invariance, at the cost of changing what the fusion layer conditions on. A demographic or otherwise privacy-bearing source would make the credit-versus-exposure trade-off the object of study rather than a complication to be avoided. Owen values [25] would apply when sources form a natural hierarchy, such as a collaborative family against a content family, and Myerson values [26] when the coalition structure is genuinely restricted, as in cascade hybrids where a downstream stage cannot operate without an upstream one. Finally, online or interleaved evaluation would test whether the segment-adaptive gains observed offline survive deployment.

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Declarations

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- **Competing interests.** The authors declare no competing interests.
- **Ethics approval.** Not applicable; the study uses publicly available secondary data with no human participants.
- **Consent to participate / for publication.** Not applicable.
- **Data availability.** All three benchmarks are derived from public corpora: MovieLens-1M [35], and the Amazon-Beauty and Amazon-Grocery subsets of the Amazon Reviews corpus [36].
- **Author contributions.** **M.L.**: conceptualization, methodology, software, formal analysis, investigation, writing, original draft. **R.N.**: methodology, validation, writing, review and editing. **M.Lz.**: supervision, conceptualization, writing, review and editing. All authors read and approved the final manuscript.

Appendix A Proofs and notation

Proposition 1. The Shapley value satisfies efficiency, so $\sum_g \varphi_g = v(\mathcal{G}) - v(\emptyset)$. By Definition 1, $v(\emptyset) = 0$. $\square$

Proposition 2. Write $D = \mathcal{G} \setminus \{g_1, g_2\}$ and let the redundancy hypothesis $v(C \cup \{g_1\}) = v(C \cup \{g_2\}) = v(C \cup \{g_1, g_2\})$ hold for every $C \subseteq D$.

Leave-one-out. Take $C = D$. Then $\mathrm{LOO}(g_1) = v(\mathcal{G}) - v(\mathcal{G} \setminus \{g_1\}) = v(D \cup \{g_1, g_2\}) - v(D \cup \{g_2\}) = 0$ by the hypothesis, and symmetrically $\mathrm{LOO}(g_2) = 0$.

Symmetry. For any $C \subseteq D$ the hypothesis gives $v(C \cup \{g_1\}) = v(C \cup \{g_2\})$, and for any $C \subseteq D$ we also have $v(C \cup \{g_2\} \cup \{g_1\}) = v(C \cup \{g_1\} \cup \{g_2\})$ trivially. Hence g_1 and g_2 are symmetric players and the symmetry axiom gives $\varphi_{g_1} = \varphi_{g_2}$.

The common value. Fix g_1 and partition the coalitions $C \subseteq \mathcal{G} \setminus \{g_1\}$ by whether they contain g_2. If $g_2 \in C$, write $C = C' \cup \{g_2\}$ with $C' \subseteq D$; the hypothesis gives $v(C' \cup \{g_2, g_1\}) = v(C' \cup \{g_2\})$, so the marginal contribution of g_1 is exactly zero and the term drops out. If $g_2 \notin C$ then $C \subseteq D$, and the hypothesis gives $v(C \cup \{g_1\}) = v(C \cup \{g_1, g_2\})$, so the marginal contribution is $v(C \cup \{g_1, g_2\}) - v(C)$. Substituting into Definition 2 and keeping the original K-player weights leaves exactly the surviving half of the sum,

$$\varphi_{g_1} = \sum_{C \subseteq D} \frac{|C|!\,(K - |C| - 1)!}{K!} \big[v(C \cup \{g_1, g_2\}) - v(C)\big],$$

which is Equation (5); the same argument with the roles exchanged gives the identical expression for φ_{g_2}, re-deriving the symmetry established above. Strict positivity follows because every weight is positive and, under the hypothesis, each bracket is non-negative whenever v is monotone, with at least one strictly positive by assumption.

What the proof does not license. The weighted sum above cannot be collapsed to the single difference $v(\mathcal{G}) - v(D)$: doing so would retain only the $C = D$ term and discard every lower-order coalition, which is legitimate only when $D = \emptyset$. Remark 1 gives an explicit three-player counterexample where the collapsed expression overstates the true value by fifty percent. $\square$

Proposition 3. The Shapley operator is linear in the characteristic function: for $v = \sum_k \alpha_k v_k$, the value of player g is $\sum_k \alpha_k \varphi_g[v_k]$. Applying this with $\alpha_u = |\mathcal{U}|^{-1}$ gives the claim. $\square$

Remark 2. Under $s \mapsto as + b$ with $a > 0$, the per-user mean and standard deviation map as $\mu \mapsto a\mu + b$ and $\sigma \mapsto a\sigma$, so the normalized value $(as + b - a\mu - b)/(a\sigma) = (s - \mu)/\sigma$ is unchanged. Every downstream quantity is a function of the normalized values alone. $\square$

Table A1 Notation.

Symbol	Meaning
$\mathcal{U}, \mathcal{I}$	user and item sets
$\mathcal{G} = \{g_1, \ldots, g_K\}$	signal sources, the players; $K = 6$
$s_g(u, i)$	score assigned to (u, i) by source g
$C \subseteq \mathcal{G}$	a coalition of sources
f_θ^C	fusion model refitted on the sources in C
$v(C)$, $v_u(C)$	characteristic function, globally and for user u
φ_g, $\varphi_g(u)$	Shapley value of source g, globally and for user u
$\bar{\varphi}_g$	normalized source share
$\mathcal{S}_1, \ldots, \mathcal{S}_M$	user segments
π	null ranker, uniform random under a fixed seed
N	candidate-set size per user
$A_u^{\mathcal{G}}$	candidate pool for user u, built once from $\mathcal{G}$
$z_{g,u,i}$	per-user, per-source z-normalized score
P_u, R	sampled training pairs for user u, and $R = |P_u|$
α	inverse regularization of the fusion layer
λ	shrinkage toward global fusion weights

Table B2 Coalition values $v(C)$, MovieLens-1M. All 64 coalitions.

Coalition	$v(C)$	Coalition	$v(C)$
∅	0.00000	CB	0.00429
CF	0.02683	EASE	0.03780
POP	0.01465	REC	0.01365
SEQ	0.05584	CB, EASE	0.03886
CB, POP	0.02319	CB, REC	0.01732
CB, SEQ	0.05883	CF, CB	0.02625
CF, EASE	0.03897	CF, POP	0.03499
CF, REC	0.03137	CF, SEQ	0.07523
POP, EASE	0.03733	POP, REC	0.01301
POP, SEQ	0.05554	REC, EASE	0.03829
REC, SEQ	0.05647	SEQ, EASE	0.05838
CB, POP, EASE	0.03856	CB, POP, REC	0.02318
CB, POP, SEQ	0.05702	CB, REC, EASE	0.03839
CB, REC, SEQ	0.05871	CB, SEQ, EASE	0.05825
CF, CB, EASE	0.03883	CF, CB, POP	0.03680
CF, CB, REC	0.03252	CF, CB, SEQ	0.07516
CF, POP, EASE	0.03765	CF, POP, REC	0.03476
CF, POP, SEQ	0.06996	CF, REC, EASE	0.03747
CF, REC, SEQ	0.07122	CF, SEQ, EASE	0.06625
POP, REC, EASE	0.03734	POP, REC, SEQ	0.05642
POP, SEQ, EASE	0.05814	REC, SEQ, EASE	0.05806
CB, POP, REC, EASE	0.03822	CB, POP, REC, SEQ	0.05709
CB, POP, SEQ, EASE	0.05872	CB, REC, SEQ, EASE	0.05854
CF, CB, POP, EASE	0.03811	CF, CB, POP, REC	0.03615
CF, CB, POP, SEQ	0.07060	CF, CB, REC, EASE	0.03774
CF, CB, REC, SEQ	0.07064	CF, CB, SEQ, EASE	0.06696
CF, POP, REC, EASE	0.03737	CF, POP, REC, SEQ	0.06966
CF, POP, SEQ, EASE	0.06824	CF, REC, SEQ, EASE	0.06655
POP, REC, SEQ, EASE	0.05819	CB, POP, REC, SEQ, EASE	0.05849
CF, CB, POP, REC, EASE	0.03817	CF, CB, POP, REC, SEQ	0.06990
CF, CB, POP, SEQ, EASE	0.06892	CF, CB, REC, SEQ, EASE	0.06651
CF, POP, REC, SEQ, EASE	0.06884	CF, CB, POP, REC, SEQ, EASE	0.06856

Appendix B Full coalition values

Tables B2, B3 and B4 give $v(C)$ for every one of the $2^6 = 64$ coalitions on each benchmark. Reporting the game in full is feasible only because it is exact and small, and we include it so that any reader can recompute the allocation of Table 9 directly from these values.

Appendix C Hyperparameter grids

Selected values are those of Table 5. The collaborative source uses 64 latent factors with regularization and learning rate both 0.01 over 100 iterations; the content source uses sublinear term frequency with a minimum document frequency of 2; the recency source uses a 30-day half-life, interpreted in ordinal sequence steps on Amazon-Beauty for the reason given in Section 5.1; the sequential source uses additive smoothing of 1; and EASE uses $\lambda_E = 250$, solved by Cholesky factorization rather than explicit inversion

Table B3 Coalition values $v(C)$, Amazon-Beauty. All 64 coalitions.

Coalition	$v(C)$	Coalition	$v(C)$
$\emptyset$	0.00000	CB	0.00692
CF	0.00739	EASE	0.02486
POP	0.00508	REC	0.00358
SEQ	0.01471	CB, EASE	0.02820
CB, POP	0.01379	CB, REC	0.01297
CB, SEQ	0.02183	CF, CB	0.01376
CF, EASE	0.02539	CF, POP	0.01062
CF, REC	0.01175	CF, SEQ	0.01943
POP, EASE	0.02374	POP, REC	0.00411
POP, SEQ	0.01826	REC, EASE	0.02370
REC, SEQ	0.01758	SEQ, EASE	0.02621
CB, POP, EASE	0.02690	CB, POP, REC	0.01339
CB, POP, SEQ	0.02223	CB, REC, EASE	0.02720
CB, REC, SEQ	0.02184	CB, SEQ, EASE	0.02945
CF, CB, EASE	0.02832	CF, CB, POP	0.01695
CF, CB, REC	0.01785	CF, CB, SEQ	0.02310
CF, POP, EASE	0.02387	CF, POP, REC	0.01122
CF, POP, SEQ	0.01982	CF, REC, EASE	0.02270
CF, REC, SEQ	0.01881	CF, SEQ, EASE	0.02630
POP, REC, EASE	0.02369	POP, REC, SEQ	0.01803
POP, SEQ, EASE	0.02563	REC, SEQ, EASE	0.02498
CB, POP, REC, EASE	0.02720	CB, POP, REC, SEQ	0.02177
CB, POP, SEQ, EASE	0.02838	CB, REC, SEQ, EASE	0.02901
CF, CB, POP, EASE	0.02569	CF, CB, POP, REC	0.01788
CF, CB, POP, SEQ	0.02374	CF, CB, REC, EASE	0.02633
CF, CB, REC, SEQ	0.02349	CF, CB, SEQ, EASE	0.02966
CF, POP, REC, EASE	0.02276	CF, POP, REC, SEQ	0.01886
CF, POP, SEQ, EASE	0.02541	CF, REC, SEQ, EASE	0.02414
POP, REC, SEQ, EASE	0.02498	CB, POP, REC, SEQ, EASE	0.02870
CF, CB, POP, REC, EASE	0.02650	CF, CB, POP, REC, SEQ	0.02377
CF, CB, POP, SEQ, EASE	0.02739	CF, CB, REC, SEQ, EASE	0.02752
CF, POP, REC, SEQ, EASE	0.02402	CF, CB, POP, REC, SEQ, EASE	0.02756

so that the 12,101-item Beauty catalogue fits within the 3 GB memory budget. The fusion layer samples 10 pairs per training user at $C = 1.0$ without an intercept. We verified that the allocation is insensitive to the pair count: raising it from 10 to 200 changes NDCG@10 on MovieLens-1M from 0.0722 to 0.0733, and the fusion objective is convex so the fit is deterministic given the sampling seed.

Appendix D Candidate-set size and the retrieval ceiling

Two design choices around candidate generation deserve their own record.

Why the candidate set is fixed across coalitions.

Regenerating candidates per coalition changes the retrieval ceiling from coalition to coalition, which breaks the comparability of $v(C)$ that the game relies on and makes

Table B4 Coalition values $v(C)$, Amazon-Grocery. All 64 coalitions.

Coalition	$v(C)$	Coalition	$v(C)$
$\emptyset$	0.00000	CB	-0.00068
CF	0.00973	EASE	0.02989
POP	0.00419	REC	0.00622
SEQ	0.01691	CB, EASE	0.02980
CB, POP	0.00419	CB, REC	0.00622
CB, SEQ	0.01689	CF, CB	0.00986
CF, EASE	0.03087	CF, POP	0.01645
CF, REC	0.01612	CF, SEQ	0.02155
POP, EASE	0.02896	POP, REC	0.00842
POP, SEQ	0.02096	REC, EASE	0.02970
REC, SEQ	0.02205	SEQ, EASE	0.03180
CB, POP, EASE	0.02896	CB, POP, REC	0.00842
CB, POP, SEQ	0.02099	CB, REC, EASE	0.02974
CB, REC, SEQ	0.02209	CB, SEQ, EASE	0.03167
CF, CB, EASE	0.03086	CF, CB, POP	0.01645
CF, CB, REC	0.01620	CF, CB, SEQ	0.02141
CF, POP, EASE	0.02906	CF, POP, REC	0.01820
CF, POP, SEQ	0.02420	CF, REC, EASE	0.02977
CF, REC, SEQ	0.02431	CF, SEQ, EASE	0.03187
POP, REC, EASE	0.02930	POP, REC, SEQ	0.02242
POP, SEQ, EASE	0.03063	REC, SEQ, EASE	0.03270
CB, POP, REC, EASE	0.02930	CB, POP, REC, SEQ	0.02241
CB, POP, SEQ, EASE	0.03070	CB, REC, SEQ, EASE	0.03264
CF, CB, POP, EASE	0.02909	CF, CB, POP, REC	0.01819
CF, CB, POP, SEQ	0.02415	CF, CB, REC, EASE	0.02974
CF, CB, REC, SEQ	0.02430	CF, CB, SEQ, EASE	0.03168
CF, POP, REC, EASE	0.02784	CF, POP, REC, SEQ	0.02610
CF, POP, SEQ, EASE	0.03092	CF, REC, SEQ, EASE	0.03279
POP, REC, SEQ, EASE	0.03214	CB, POP, REC, SEQ, EASE	0.03213
CF, CB, POP, REC, EASE	0.02799	CF, CB, POP, REC, SEQ	0.02602
CF, CB, POP, SEQ, EASE	0.03094	CF, CB, REC, SEQ, EASE	0.03268
CF, POP, REC, SEQ, EASE	0.03076	CF, CB, POP, REC, SEQ, EASE	0.03076

v non-monotone: a coalition could score worse than a subset of itself purely because it retrieved a harder candidate pool. We pre-committed to the fixed-candidate design in Section 4.3 for this reason. The cost is stated honestly in Section 6.4: masking a source at fusion time is not the same as never having built it, because every coalition inherits a pool that the grand coalition helped retrieve.

How much the fixed pool changes the allocation.

The argument above says why the pool is held fixed; it does not say what that choice costs. Because the objection is about magnitude rather than principle, we measured it. Table D5 recomputes the entire game with candidates *regenerated* from each coalition's own sources, so that a source excluded from a coalition no longer helps build the pool that coalition is scored on. This is the end-to-end quantity the fixed-pool design cannot see, and it costs roughly $2^K/K$ times more scoring work, which is why it is a check rather than the default protocol.

The two protocols coincide exactly at the grand coalition, as they must, since the regenerated pool for $\mathcal{G}$ *is* the fixed pool; $v(\mathcal{G})$ is identical to machine precision on all three benchmarks, which is a useful correctness check on the reimplementation. Away from the grand coalition they differ, and the retrieval ceiling now moves with the coalition: on MovieLens-1M it ranges from 0.596 for the weakest single source to 0.885 at the grand coalition, on Amazon-Beauty from 0.238 to 0.605, and on Amazon-Grocery from 0.488 to 0.637. As anticipated, v is no longer monotone under this protocol on any of the three datasets.

The effect on the allocation is nonetheless small. Comparing the two protocols on the same seed, so that only the protocol differs, the largest change in any source's normalized share is 1.2 percentage points on MovieLens-1M, 0.6 on Amazon-Beauty and 1.0 on Amazon-Grocery, so no share moves by more than 1.2 points anywhere, and the source ordering is preserved on all three benchmarks. For scale, this is smaller than the seed-to-seed variation of the fixed protocol itself, which Table 16 reports at up to about two percentage points. The largest single movement is EASE on MovieLens-1M, which loses 1.2 points once the other sources must retrieve their own candidates; this is the direction one expects, since EASE is an item–item model whose retrieved pool overlaps heavily with the collaborative source it is being separated from.

We draw a bounded conclusion from this. It does *not* show that fusion-stage credit equals end-to-end source value, and we do not claim it does. What it shows is that on these three benchmarks the retrieval-stage contribution of a source is not large enough to reorder the allocation or to move any share by more than about one point, so the ranking of sources by credit, which is what Section 6.1 actually uses, is robust to the design choice. A source whose value lies mostly in retrieval rather than in ranking would break this, and we would expect such a source to exist in a catalogue far larger than ours.

Why N was raised.

Table D6 reports the measured ceiling as a function of N. At the conventional $N = 200$ the ceiling is 0.504 on MovieLens-1M and 0.160 on Amazon-Beauty, meaning that on the sparse benchmark more than four fifths of users could not be served correctly by *any* coalition. Since NDCG is bounded above by this quantity, attribution computed at $N = 200$ would describe a system whose behaviour is dominated by retrieval failure rather than by fusion. We therefore raised N to 1000 on MovieLens-1M and 3000 on both Amazon benchmarks, the smallest round values that clear a 0.5 ceiling. The ceiling applies uniformly across coalitions, so this rescales all reported NDCG values without biasing the allocation between sources.

Appendix E Per-segment attribution in full

Table 13 in the main text gives the per-segment allocation for MovieLens-1M. Tables E7 and E8 complete the picture for the two sparse benchmarks. Segments are activity quartiles fitted on training users only and applied to held-out users by the frozen segmenter, as described in Section 4.7.

Table D5 Effect of regenerating candidates per coalition. "Fixed" is the protocol used throughout the paper, in which one pool is built from the grand coalition and shared; "Regen." rebuilds the pool from each coalition's own sources. Values are normalized Shapley shares in percent, and Δ is given in percentage points. **Both columns are computed on seed 0 alone**, not on the five-seed mean used in Table 9, because the regenerated protocol costs roughly $2^K/K$ times more scoring work than the fixed one and a five-seed sweep was not affordable. The "Fixed" column therefore differs from Table 9 by the usual seed-to-seed variation of up to about two percentage points, which Table 16 quantifies separately. The comparison that matters here is *within* a row: both columns share a seed, so Δ isolates the protocol change. The two protocols agree exactly at the grand coalition by construction.

	MovieLens-1M			Amazon-Beauty			Amazon-Grocery		
Source	Fixed	Regen.	Δ	Fixed	Regen.	Δ	Fixed	Regen.	Δ
CF	19.3	19.7	+0.4	8.0	8.1	+0.0	12.1	11.6	-0.5
CB	2.4	2.7	+0.3	17.8	17.2	-0.6	-0.4	-0.6	-0.2
POP	5.7	6.1	+0.4	4.2	4.5	+0.2	3.5	3.2	-0.2
REC	3.9	4.0	+0.2	2.9	3.0	+0.2	7.7	8.7	+1.0
SEQ	53.1	53.0	-0.1	23.7	23.5	-0.2	26.8	26.7	-0.1
EASE	15.6	14.4	-1.2	43.4	43.8	+0.4	50.3	50.4	+0.1

Table D6 Candidate recall ceiling as a function of the candidate-set size N, measured before any attribution was computed. Bold entries are the selected operating points and correspond to the final column of Table 6; the remaining entries are from the N-sweep run at selection time. Dashes mark values not measured, the sweep having already cleared the threshold. The non-bold entries were measured during model selection and are reported as recorded; only the selected operating points were recomputed for the final run, which is why they carry more decimals.

Dataset	$N = 200$	$N = 500$	$N = 1000$	$N = 2000$	$N = 3000$
MovieLens-1M	0.504	0.725	**0.8847**	—	—
Amazon-Beauty	0.160	0.245	0.346	0.500	**0.6050**
Amazon-Grocery	0.167	—	0.383	0.516	**0.6368**

The attribution-based segmentation is deliberately not tabulated. Its agreement with the behavioural segmentation is indistinguishable from chance on all three benchmarks (adjusted Rand = 0.0029, −0.0018, −0.0009), so the claim is withdrawn in Section 5.5 rather than illustrated here.

Appendix F Attribution baselines on the sparse benchmarks

Table 12 in the main text compares the attribution methods on MovieLens-1M. Tables F9 and F10 repeat that comparison on the two sparse benchmarks, where the leave-one-out shortfall is larger still.

Table E7 Source shares (%) by behavioural segment, Amazon-Beauty. Entries marked † are negative, so per Section 4.4 they are not interpretable as percentage shares; the underlying raw φ_g values are the meaningful quantity there.

Segment	n	CF	CB	POP	REC	SEQ	EASE
Q1 (coldest)	1289	6.1	22.3	4.6	1.5	17.8	47.7
Q2	1252	6.6	20.3	4.4	2.9	21.8	44.0
Q3	587	12.1	6.9	8.7	13.1	15.5	43.7
Q4 (heaviest)	872	11.8	11.9	-0.1†	-1.9†	45.3	33.0

Permutation test on between-segment variance: observed = 0.0258, $p = 0.0385$
Attribution-vs-behavioural agreement (adjusted Rand): -0.0018

Table E8 Source shares (%) by behavioural segment, Amazon-Grocery. Entries marked † are negative, so per Section 4.4 they are not interpretable as percentage shares; the underlying raw φ_g values are the meaningful quantity there.

Segment	n	CF	CB	POP	REC	SEQ	EASE
Q1 (coldest)	1186	10.2	-0.1†	4.5	-3.0†	27.2	61.4
Q2	1142	16.1	-0.5†	6.4	6.2	18.5	53.2
Q3	822	12.5	-0.6†	3.6	4.7	12.6	67.3
Q4 (heaviest)	850	5.9	-0.5†	-7.2†	43.9	71.3	-13.3†

Permutation test on between-segment variance: observed = 0.1953, $p = 0.0005$
Attribution-vs-behavioural agreement (adjusted Rand): -0.0009

Table F9 Attribution methods compared, Amazon-Beauty, on the five-seed-mean basis used throughout.

Source	φ_g	LOO	Fwd. sel.	Perm.	Max. redund.
CF	0.00249	-0.00072	0.00050	0.00424	0.218
CB	0.00477	0.00358	0.00340	0.00653	0.193
POP	0.00118	0.00006	-0.00012	0.00012	0.955
REC	0.00103	0.00013	-0.00101	0.00596	0.955
SEQ	0.00598	0.00090	0.00029	0.00128	0.097
EASE	0.01267	0.00535	0.02506	0.01170	0.173
Sum	0.02811	0.00931	—	—	—
$v(\mathcal{G})$			0.02811		
LOO recovers			33.1% of $v(\mathcal{G})$		

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Table F10 Attribution methods compared, Amazon-Grocery, on the five-seed-mean basis used throughout.

Source	φ_g	LOO	Fwd. sel.	Perm.	Max. redund.
CF	0.00320	-0.00131	0.00055	0.00135	0.176
CB	-0.00018	-0.00001	-0.00003	-0.00001	0.061
POP	0.00050	-0.00268	-0.00268	-0.00147	0.292
REC	0.00295	0.00068	0.00180	0.00157	0.292
SEQ	0.00845	0.00305	0.00185	0.00341	0.096
EASE	0.01498	0.00498	0.02842	0.01773	0.165
Sum	0.02991	0.00471	—	—	—
$v(\mathcal{G})$			0.02991		
LOO recovers			15.7% of $v(\mathcal{G})$		

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